Plant tissue (apple and potato parenchyma, fruit skins, seaweed, nuts)

Blahovec, J. (1988). Mechanical properties of some plant materials. *J. Mater. Sci.*, *23*(10), 3588–3593.

Gibson, L.J., & Ashby, M.F. (1997). *Cellular Solids: Structure and Properties* (2nd ed.). Cambridge University Press.

Greenberg, A.R., Mehling, A., Lee, M., & Bock, J.H. (1989). Tensile behavior of grass. *J. Mater. Sci.*, *24*(7), 2549–2554.

Jennings, J.S., & Macmillan, N.H. (1986). A tough nut to crack. *J. Mater. Sci.*, *21*(5), 1517–1524.

Venkataswamy, M.A., Pillai, C.K.S., Prasad, V.S., & Satyanarayana, K.G. (1987). Effect of weathering on the mechanical properties of midribs of coconut leaves. *J. Mater. Sci.*, *22*(9), 3167–3172.

Vincent, J.F.V. (1989). Relationship between density and stiffness of apple flesh. *J. Sci. Food. Agr.*, *47*(4), 443–462.

Vincent, J.F.V. (1990). Fracture properties of plants. *Adv. Bot. Res.*, *17*, 235–287.

Wang, C.H., Zhang, L.C., & Mai, Y.W. (1994). Deformation and fracture of macadamia nuts. 1. Deformation analysis of nut-in-shell. *Int. J. Fracture*, *69*(1), 51–65.

Wang, C.H., & Mai, Y.W. (1994). Deformation and fracture of macadamia nuts. 2. Microstructure and fracture mechanics analysis of nutshell. *Int. J. Fracture*, *69*(1), 67–85.

Cellulose, cotton, flax, hemp, jute, ramie

Currey, J.D., Wainwright, S.A., & Biggs, W.D. (1982). *Mechanical design in organisms*. Princeton University Press.

Gibson, L.J., & Ashby, M.F. (1997). *Cellular Solids: Structure and Properties* (2nd ed.). Cambridge University Press.

Vincent, J.F.V. (1990). *Structural biomaterials* (revised edition). Princeton University Press.

Wainwright, S.A, Gosline, J.M., & Biggs, W.D. (1992). *Mechanical design in organisms*. Princeton University Press

Wainwright, S.A. (1980). Adaptive materials: a view from the organism. In J.F.V. Vincent, & J.D. Currey (Eds.), The mechanical properties of biological materials. *Proceedings of the Symposia of the Society for Experimental Biology*, *34*, 483–453. Cambridge University Press for the Society for Experimental Biology.

Silk

Calvert, P. (1989). Natural polymers—spinning ties that bind. *Nature*, *340*, 266.

Denny, M. (1980). Silks—their properties and functions. In J.F.V. Vincent & J.D. Currey (Eds.), The mechanical properties of biological materials. *Proceedings of the Symposia of the Society for Experimental Biology*, *34*, 247–272. Cambridge University Press for the Society for Experimental Biology.

Gosline, J.M., Demont, M.E., & Denny, M.W. (1986). The structure and properties of spider silk. *Endeavour*, *10*(1), 37–43.

Elastin and resilin

Goseline, J.M. (1980). The elastic properties of rubber-like proteins and highly extensible tissues. In J.F.V. Vincent & J.D. Currey (Eds.), The mechanical properties of biological

materials. *Proceedings of the Symposia of the Society for Experimental Biology, 34*, 331–357. Cambridge University Press for the Society for Experimental Biology.

Oxlund, H., Manschot, J., & Viidik, A. (1988). The role of elastin in the mechanical properties of skin. *J. Biomech., 21*(3), 213–218.

Collagen, ligament, tendon

Andersen, K.L., Pedersen, E.H., & Melsen, B. (1991). Material parameters and stress profiles within the periodontal ligament. *Am. J. Orthod. Dentofac. Orthop., 99*(5), 427–440.

Bennett, M.B., Ker, R.F., Dimery, N.J., & Alexander, R.M. (1986). Mechanical properties of various mammalian tendons. *J. Zool., 209*(4), 537–548.

Bosch, U., Decker, B., Kasperczyk, W., Nerlich, A., Oestern, H.J., & Tscherne, H. (1992). The relationship of mechanical properties to morphology in patellar tendon autografts after posterior cruciate ligament replacement in sheep. *J. Biomech., 25*(8), 821–830.

Butler, D.L., Kay, M.D., & Stouffer, D.C. (1986). Comparison of material properties in fascicle-bone units from human patellar tendon and knee ligaments. *J. Biomech., 19*(6), 425–432.

Grieshaber, F.A., & Faust, U. (1992). Mechanische kenngrößen von biologischem weichgewebe. *Biomed. Technik, 37*(12), 278–286.

Kato, Y.P., Christiansen, D.L., Hahn R.A., Shieh, S.J., Goldstein, J.D., & Silver, F.H. (1989). Mechanical properties of collagen fibres: A comparison of reconstituted and rat tendon fibres. *Biomat., 10*, 38–41.

Ker, R.F. (1981). Dynamic tensile properties of the plantaris tendon of sheep (*Ovis aries*). *J. Exp. Biol., 93*(Aug.), 283–302.

Ker, R.F., Alexander, R.M., & Bennett, M.B. (1988). Why are mammalian tendons so thick. *J. Zool., 216*(2), 309–324.

Park, J.B., & Lakes, R.S. (1992). *Biomaterials* (2nd ed.). Plenum Press.

Rogers, G.J., Milthorpe, B.K., Muratore, A., & Schindhelm, K. (1990). Measurement of the mechanical properties of the ovine anterior cruciate ligament bone complex—a basis for prosthetic evaluation. *Biomaterials, 11*(2), 89–96.

Silver, F.H., Kato, Y.P., Ohno, M., & Wasserman, A.J. (1992). Analysis of mammalian connective-tissue—relationship between hierarchical structures and mechanical properties. *J. Long Term Effects Med. Impl., 2*(2–3), 165–198.

Vincent, J.F.V. (1990). *Structural Biomaterials* (revised edition). Princeton University Press.

Wainwright, S. (1980). Adaptive materials: A view from the organism. In J.F.V. Vincent & J.D. Currey (Eds.), The mechanical properties of biological materials. *Proceedings of the Symposia of the Society for Experimental Biology, 34*, 483–453. Cambridge University Press for the Society for Experimental Biology.

Wang, X.T., & Ker, R.F. (1995a). Creep-rupture of wallaby tail tendons. *J. Exp. Biol., 198*(3), 831–845.

Wang, X.T., Ker, R.F., & Alexander, R.M. (1995b). Fatigue-rupture of wallaby tail tendons. *J. Exp. Biol., 198*(3), 847–852.

Muscle

Dobrunz, L.E., Pelletier, D.G., & McMahon, T.A. (1990). Muscle-stiffness measured under conditions simulating natural sound production. *Biophys. J., 58*(2), 557–565.

Yamada, H. (1970). *Strength of biological materials.* Williams & Wilkins.

Skin

Bauer, A.M., Russell, A.P., & Shadwick, R.E. (1989). Mechanical properties and morphological correlates of fragile skin in gekkonid lizards. *J. Exp. Biol.*, *145*, 79–102.

Manschot, J.F.M., & Brakkee, A.J.M. (1986a). The measurement and modeling of the mechanical properties of human skin invivo. 1. The measurement. *J. Biomech.*, *19*(7), 511–515.

Manschot, J.F.M., & Brakkee, A.J.M. (1986b). The measurement and modeling of the mechanical properties of human skin invivo. 2. The model. *J. Biomech.*, *19*(7), 517–521.

Oxlund, H., Manschot J., & Viidik, A. (1988). The role of elastin in the mechanical properties of skin. *J. Biomech.*, *21*(3), 213–218.

Silver, F.H., Kato, Y.P., Ohno, M., & Wasserman, A.J. (1992). Analysis of mammalian connective tissue—relationship between hierarchical structures and mechanical properties. *J. Long Term Effects Med. Impl.*, *2*(2–3), 165–198.

Swartz, S.M., Groves, M.S., Kim, H.D., & Walsh, W.R. (1996). Mechanical properties of bat wing membrane skin. *J. Zool.*, *239*(2), 357–378.

Leather

Attenburrow, G.E., & Wright, D.M. (1994). Studies of the mechanical behavior of partially processed leather. *J. Am. Leather Chem. Assoc.*, *89*(12), 391–402.

Kronick, P., & Maleef, B. (1992). Nondestructive failure testing of bovine leather by acoustic-emission. *J. Am. Leather Chem. Assoc.*, *87*(7), 259–265.

Lin, J., & Hayhurst, D.R. (1993a). The development of a bi-axial tension test facility and its use to establish constitutive equations for leather. *Eur. J. Mech. A—Solids*, *12*(4), 493–507.

Lin, J., & Hayhurst, D.R. (1993b). Constitutive equations for multiaxial straining of leather under uniaxial stress. *Eur. J. Mech. A—Solids*, *12*(4,), 471–492.

Cartilage

Rains, J.K., Bert, J.L., Roberts, C.R., & Pare, P.D. (1992). Mechanical properties of human tracheal cartilage. *J. Appl. Physiol.*, *72*(1), 219–225.

Silver, F.H., Kato, Y.P., Ohno, M., & Wasserman, A.J. (1992). Analysis of mammalian connective tissue—relationship between hierarchical structures and mechanical properties. *J. Long Term Effects Med. Impl.*, *2*(2-3), 165–198.

Swanson, S. (1980). The elastic properties of rubber-like protein and highly extensible tissue. In J.F.V. Vincent & J.D. Currey (Eds.), The mechanical properties of biological materials. *Proceedings of the Symposia of the Society for Experimental Biology*, *34*, 377–395. Cambridge University Press for the Society for Experimental Biology.

Bone (including antler)

Ashman, R.B., Corin, J.D., & Turner, C.H. (1987). Elastic properties of cancellous bone—measurement by an ultrasonic technique. *J. Biomech.*, *20*(10), 979–986.

Ashman, R.B., & Rho, J.Y. (1988). Elastic modulus of trabecular bone material. *J. Biomech.*, *21*(3), 177–181.

Currey, J.D. (1984). *The mechanical adaptations of bones*. Princeton University Press.

Currey, J.D. (1988). The effects of drying and re-wetting on some mechanical properties of cortical bone. *J. Biomech.*, *21*(5), 439–441.

Currey, J.D. (1990). Physical characteristics affecting the tensile failure properties of compact bone. *J. Biomech.*, *23*(8), 837–844.

Dickenson, R.P., Hutton, W.C., & Stott, J.R.R. (1981). The mechanical properties of bone in osteoporosis. *J. Bone Joint Surg.—Brit. Vol.*, *63*(2), 233–238.

Gibson, L.J. (1985). The mechanical behavior of cancellous bone. *J. Biomech.*, *18*(5), 317 et seq.

Goldstein, S.A. (1987). The mechanical properties of trabecular bone—dependence on anatomic location and function. *J. Biomech.*, *20*(11–12), 1055–1061.

Swanson, S. (1980). The elastic properties of rubber-like protein and highly extensible tissue. In J.F.V. Vincent & J.D. Currey (Eds.), The mechanical properties of biological materials. *Proceedings of the Symposia of the Society for Experimental Biology, 34*, 377–395. Cambridge University Press for the Society for Experimental Biology.

Kitchener, A. (1991). The evolution and mechanical design of horns and antlers. In J. Rayner & R. Wootton (Eds.), *Society for experimental biology seminar series: no. 36. Biomechanics in evolution* (pp. 229–253). Cambridge University Press.

Linde, F., & Hvid, I. (1987). Stiffness behavior of trabecular bone specimens. *J. Biomech.*, *20*(1), 83–89.

Lotz, J.C., Gerhart, T.N., & Hayes, W.C. (1991). Mechanical properties of metaphyseal bone in the proximal femur. *J. Biomech.*, *24*(5), 317–329.

Moyle, D.D., & Gavens, A.J. (1986). Fracture properties of bovine tibial bone. *J. Biomech.*, *19* (11), 919–927.

Moyle, D.D., & Walker, M.W. (1986). The effects of a calcium deficient diet on the mechanical-properties and morphology of goose bone. *J. Biomech.*, *19*(8), 613–625.

Sharp, D.J., Tanner, K.E., & Bonfield, W. (1990). Measurement of the density of trabecular bone. *J. Biomech.*, *23*(8), 853–857.

Sun, J.J., & Geng, J. (1987). A study of Haversian systems. *J. Biomech.*, *20*(8), 815.

Watkins, M. (1987). *The development of a tough artificial composite based on antler bone*. PhD thesis, University of Reading.

Dentine, enamel

Brear, K., Currey, J.D., Pond, C.M., & Ramsay, M.A. (1990). The mechanical properties of the dentin and cement of the tusk of the narwhal *Monodon monoceros* compared with those of other mineralized tissues. *Arch. Oral Biol.*, *35*(8), 615–621.

Coral

Chamberlain, J. (1978). Mechanical properties of coral skeleton: compressive strength and its adaptive significance. *Paleobiol.*, *4*, 419–435.

Kim, K., Goldberg, W.M., & Taylor, G.T. (1992). Architectural and mechanical properties of the black coral skeleton (*Coelenterata*, *Antipatharia*)—a comparison of 2 species. *Biol. Bull.*, *182*(2), 195–209.

Scott, P.J.B., & Risk, M.J. (1988). The effect of *Lithophaga* (*Bivalvia*, *Mytilidae*) boreholes on the strength of the coral *Porites lobata*. *Coral Reefs*, *7*(3), 145–151.

Vosburgh, F. (1982). *Acropora reticulata*—structure, mechanics and ecology of a reef coral. *P. Roy. Soc. Lond. B*, *214*(1197), 481–499.

Alpha-keratins

Bertram, J.E.A., & Gosline, J.M. (1986). Fracture toughness design in horse hoof keratin. *J. Exp. Biol.*, *125*, 29–47.

Bertram, J.E.A., & Gosline, J.M. (1987). Functional design of horse hoof keratin—the modulation of mechanical properties through hydration effects. *J. Exp. Biol.*, *130*, 121–136.

Fraser, R., & Macrae, T. (1980). Molecular structure and mechanical properties of keratins. In J.F.V. Vincent & J.D. Currey (Eds.), The mechanical properties of biological materials. *Proceedings of the Symposia of the Society for Experimental Biology, 34*, 211–246. Cambridge University Press for the Society for Experimental Biology.

Kitchener, A. (1991). The evolution and mechanical design of horns and antlers. In J. Rayner & R, Wootton (Eds.), *Society for experimental biology seminar series: no.36. Biomechanics in evolution* (pp. 229–253). Cambridge University Press.

Insect cuticle

Alexander, D.E., Blodig, J., & Hsieh, S.Y. (1995). Relationship between function and mechanical properties of the pleopods of isopod crustaceans. *Invertebrate Biol., 114*(2), 169–179.

Gunderson, S., & Whitney, J. (1992). Insect cuticle microstructure and its application to advanced composites. *Biomimetics, 1*(2), 177–197.

Vincent, J. (1980). Insect cuticle: A paradigm for natural composites. In J.F.V. Vincent & J.D. Currey (Eds.), The mechanical properties of biological materials. *Proceedings of the Symposia of the Society for Experimental Biology, 34*, 183–210. Cambridge University Press for the Society for Experimental Biology.

Mollusk shell

Jackson, A.P., Vincent, J.F.V., & Turner, R.M. (1988). The mechanical design of nacre. *P. Roy. Soc. Lond. B, 234*(1277), 415 et seq.

Jackson, A.P., Vincent, J.F.V., & Turner, R.M. (1990). Comparison of nacre with other ceramic composites. *J. Mater. Sci., 25*(7), 3173–3178.

D2.12 Measures of environmental impact

Aggregain (2007). The Waste and Resources Action Program (WRAP). *www.wrap.org.uk*
Data and an Excel-based tool to calculate energy and carbon footprint of recycled road-bed materials.

AMC (2006). Australian Magnesium Corporation, *www.aph.gov.au/house/committee/environ/greenhse/gasrpt/Sub65-dk.pdf*

APME (1997 ,1998, 1999, 2000). Eco–profiles of the European plastics industry. Association of Plastics Manufacturers in Europe. *www.plasticseurope.org*

Ashby, M.F. (2009). *Materials and the environment.* Butterworth-Heinemann.

BCA (2007). *A carbon strategy for the cement industry.* British Cement Association, *www.cementindustry.co.uk*

Boustead Model 5 (2007). Boustead Consulting. *www.boustead-consulting.co.uk*
An established life-cycle assessment tool.

Boustead, I. (1999–2006). APME Association of Plastics Manufacturers in Europe, Report series.

British Metals Recycling Association, *www.britmetrec.org.uk*

Building Research Establishment (2006). BRE Environmental Profiles database. BRE Environment Division.

BUWAL (1996). Bundesamt für umwelt, wald und landwirtschaft, Environmental series No. 250, *Life Cycle Inventories for Packaging, Vols. I and II.*

Chapman, P.F., & Roberts, F. (1993). *Metals resources and energy.* Butterworths.

Chemlink Australasia (1997). *www.chemlink.com.au/mag&oxide.htm*

ELCD (2008). *http://lca.jrc.ec.europa.eu*
A high-quality Life Cycle Inventory (LCI) core data sets of this first version of the Commission's European Reference Life Cycle Data System (ELCD).

Energy Information Association (2008). *www.eia.doe.gov*
Official energy statistics from the U.S. Government.

European Aluminium Association (2000). *www.eaa.net*

Green Design Initiative, *www.ce.cmu.edu/GreenDesign*

GREET (2007). Argonne National Laboratory and the U.S. Department of Transport, *www.transportation.anl.gov*
Software for analyzing vehicle energy use and emissions.

Hammond, G., & Jones, C. (2006). Inventory of carbon and energy (ICE). Department of Mechanical Engineering, University of Bath.

IDEMAT, Environmental Materials Database, *www.idemat.nl*

International Aluminum Institute (2000). Life cycle inventory of the worldwide aluminum industry Part 1—automotive, *www.world-aluminum.org*

Kemna, R., van Elburg, M., Li, W., & van Holsteijn, R. (2005). Methodology study of eco-design of energy-using products. Van Holsteijn en Kemna BV (VHK).

Kennedy, J. (1997). Energy minimisation in road construction and maintenance. Best Practice Report for the U.K. Department of the Environment.

Lafarge Cement, Lafarge Cement UK, *www.lafarge.com*

Lawson, B. (1996). Building materials, energy and the environment: Towards ecologically sustainable development. RAIA, Canberra, *www.climatechange.gov.au*

Lime Technology (2007). *www.limetechnology.co.uk*

MEEUP Methodology Report, final (2005).VHK. *www.pre.nl*
A report by the Dutch consultancy VHK commissioned by the European Union, detailing its implementation of an LCA tool designed to meet the EU Energy-Using Products directive.

Pentron, *www.pentron.com*

Pilz, H., Schweighofer, J., & Kletzer, E. (2005). The contribution of plastic products to resource efficiency. *Gesellschaft fur umfassende analysen (GUA)*, Vienna.

Schlesinger, M.E. (2007). *Aluminum recycling*. CRC Press.

Stiller, H. (1999). *Material intensity of advanced composite materials*. Wuppertal Institut fur Klima, Umvelt, Energie.

Sustainable concrete (2008). *www.sustainableconcrete.org.uk/*
A website representing U.K. concrete producers carrying useful information about carbon footprint.

Szargut, J., Morris, D.R., Steward, F.R. (1988). *Energy analysis of thermal chemical and metallurgical processes*. Hemisphere.

Szokolay, S.V. (1980). *Environmental science handbook: For architects and builders*. Construction Press.

The Nickel Institute North America (2007). *www.nickelinstitute.org.*

Waste Online (2008). *www.wasteonline.org.uk/resources/InformationSheets/Plastics.htm*

D.3 INFORMATION FOR MANUFACTURING PROCESSES

Alexander, J.M., Brewer, R.C., & Rowe, G.W. (1987). *Manufacturing technology, Vol. 2: Engineering processes*. Ellis Norwood.

Bralla, J.G. (1986). *Handbook of product design for manufacturing*. McGraw-Hill.

Budinski, K.G., & Budinski, M.K. (2010). *Engineering materials, properties and selection* (9th ed.). Prentice Hall.
A well-established materials text that deals well with both material properties and processes.

Waterman, N.A., & Ashby, M.F. (Eds.) (1996). *Materials selector*. Chapman and Hall.

Dieter, G.E. (1999). *Engineering design, a materials and processing approach* (3rd ed.). McGraw-Hill.
A well-balanced and respected text focusing on the place of materials and processing in technical design.

Kalpakjian, S. (1984). *Manufacturing processes for engineering materials*. Addison Wesley.

Lascoe, O.D. (1989). *Handbook of fabrication processes*. ASM International.

Schey, J.A. (1977). *Introduction to manufacturing processes*. McGraw-Hill.

Suh, N.P. (1990). *The principles of design*. Oxford University Press.

Software

CES: Cambridge Engineering Selector (2010). *www.grantadesign.com*.
Comprehensive selection system for all classes of materials and manufacturing processes. Variety of optional reference data sources, connects directly to www.matdata.net. Windows and web format. Modest price.

Internet sources

Cast Metals Federation, *www.castmetalsfederation.com*

Castings Technologies International, *www.castingstechnology.com*

Confederation of British Metalforming, *www.britishmetalforming.com*

National Center for Excellence in Metalworking Technology, *www.ncemt.ctc.com*

PERA, *www.pera.com*

The British Manufacturing Plant Constructors' Association, *www.bmpca.org.uk*

TWI (The Welding Institute), *www.twi.co.uk*

D.4 DATABASES AND EXPERT SYSTEMS AS SOFTWARE

The number and quality of computer-based materials information systems is growing rapidly. A selection of these, with comment and source, is given here. The databases are listed in alphabetical order.

Active Library on Corrosion. ASM International.
PC format requiring CD-ROM drive. Graphical, numerical, and textual information on corrosion of metals. Modest price.

ASM Handbooks Online. ASM International, *www.asminternational.org*
Annual license fee.

Alloycenter. ASM International, *www.asminternational.org*
Annual license fee.

ALUSELECT P1.0 (1992). *Engineering property data for wrought aluminum alloys*. European Aluminum Association.
PC format, DOS environment. Mechanical, thermal, electrical, and environmental properties of wrought aluminum alloys. Inexpensive.

CAMPUS: Computer Aided Material Preselection by Uniform Standards (1995), *www.campusplastics.com*
Polymer data from approximately 30 suppliers, measured to set standards.

CETIM-EQUIST II: Centre Technique des Industries Mécaniques (1997). Senlis: CETIM.
PC format, DOS environment. Compositions and designations of steels.

CETIM-Matériaux: Centre Technique des Industries Mécaniques (1997). Senlis: CETIM.
Online system. Compositions and mechanical properties of materials.

CETIM-SICLOP: Centre Technique des Industries Mécaniques (1997). Senlis: CETIM.
Online system. Mechanical properties of steels.

Cambridge Engineering Selector (2010), *www.grantadesign.com*
Comprehensive selection system for all classes of materials and manufacturing processes. Variety of optional reference data sources; connects directly to www.matdata.net. Windows and web format. Modest price.

CUTDATA: Machining Data System. Cincinnati: Metcut Research Associates, Manufacturing Technology Division.
A PC-based system that guides the choice of machining conditions: tool materials, geometries, feed rates, cutting speeds, and so forth. Modest price.

EASel: Engineering Adhesives Selector Program (1986). The Design Centre.
A knowledge-based program to select industrial adhesives for joining surfaces. Modest price.

SF-CD (replacing ELBASE): Metal Finishing/Surface Treatment Technology (1992). Metal Finishing Information Services.
Comprehensive information on published data related to surface treatment technology. Regularly updated. Modest price.

Material Data Network, *www.matdata.net*
The Material Data Network provides integrated access to a variety of quality information source, from ASM International, Granta Design, TWI, NPL, UKSteel, Matweb, IDES, etc.

PAL II: Permabond Adhesives Locator (1996). Eastleigh: Permabond.
A knowledge-based PC system for adhesive selection among Permabond adhesives. An impressive example of an expert system that works. Modest price.

PROSPECT: Version 1.1 (1995). Oxford Forestry Institute, Department of Plant Sciences, Oxford University.
A database of the properties of tropical woods of interest to a wood user; includes information about uses, workability, treatments, origins.

OPS (Optimal Polymer Selector), *www.grantadesign.com*
Integrates Granta Design's generic PolymerUniverse database, with chemical resistance data from RAPRA and grade-specific data for approximately 6000 polymers from CAMPUS, to produce the only comprehensive polymer selection system available.

teCal, Steel Heat-Treatment Calculations. ASM International.
Computes the properties resulting from defined heat treatments of low-alloy steels, using the composition as input. Modest price.

SOFINE PLASTICS (1997). Villeurbanne Cedex: Société CERAP.
Database of polymer properties. Environment and price unknown.

TAPP 2.0, Thermochemical and Physical Properties (1994). ES Microware.
+PC format, CD-ROM, Windows environment. A database of thermochemical and physical properties of solids, liquids, and gases, including phase diagrams neatly packaged with good user manual. Modest price.

UNSearch: Unified Metals and Alloys Composition Search. Materials Park: ASTM.
PC format, DOS environment. A database of information about composition, U.S. designation, and specification of common metals and alloys. Modest price.

Wegst, C.W. (1997). *Stahlschlüssel (Key to Steel)* (17th ed.). Verlag Stahlschlüssel Wegst GmbH.
CD-ROM, PC format. Excellent coverage of European products and manufacturers.

Woods of the World (1994). Tree Talk, Inc.
A CD-ROM of woods, with illustrations of structure, information about uses, origins, habitat, etc. PC format, requiring CD drive; Windows environment.

D.5 ADDITIONAL USEFUL INTERNET SITES

Metals prices and economic reports

American Metal Market, *www.amm.com*

Business Communications Company, *www.bccresearch.com*

Daily Economic Indicators, *www.bullion.org.za*

Iron and Steel Statistics Bureau, *www.issb.co.uk*

Kitco Inc Gold & Precious Metal Prices, *www.kitco.com/market*

London Metal Exchange, *www.lme.co.uk*

Metal Bulletin, *www.metalbulletin.plc.uk*

Metal Powder Report, *www.metal-powder.net*

Metallurgia, *www.metallurgia-italiana.net*

Minerals Information, *http://minerals.usgs.gov/minerals*

Roskill Reports *www.roskill.com*

The Precious Metal and Gem Connection, *www.thebulliondesk.com*

Supplier registers, government organizations, professional societies and trade associations, and standards organizations

Supplier registers

IndustryLink, *www.industrylink.com*

Kellysearch, *www.kellysearch.com*

Thomas Register, *www.thomasregister.com*

Government organizations

Commonwealth Scientific and Industrial Research Organization (Australia), *www.csiro.au*

National Academies Press, *www.nap.edu*

National Institute of Standards and Technology (U.S.A.), *www.nist.gov*

The Fraunhofer Institute, *www.fhg.de*

Professional societies and trade associations

Aluminium Federation, *www.alfed.org.uk*

ASM International, *www.asminternational.org*

ASME International, *www.asme.org*

British Constructional Steel Work Association, *www.steelconstruction.org*

Cast Metals Federation, *www.castmetalsfederation.com*

Community Trade Union, *www.community-tu.org*

Confederation of British Metalforming, *www.britishmetalforming.com*

Copper Development Association, *www.cda.org.uk*

Institute of Cast Metal Engineers, *www.icme.org.uk*

Institute of Spring Technology, *www.ist.org.uk*

International Aluminium Institute, *www.world-aluminum.org*

International Council on Mining and Metals, *www.icmm.com*

International Titanium Association, *www.titanium.org*
International Zinc Association, *www.iza.com*
Lead Development Association International, *www.ldaint.org*
Materials Information Service, *www.iom3.org*
National Association. of Steel Stock Holders, *www.nass.org.uk*
Nickel Institute, *www.nidi.org*
Pentron, *www.pentron.com*
Society of Automotive Engineers (SAE), *www.sae.org*
Steel Manufacturers Association, *www.steelnet.org*
Steel Construction Institute, *www.steel-sci.org*
The American Ceramics Society, *http://ceramics.org*
The British Manufacturing Plant Constructors'Association, *www.bmpca.org.uk*
The Institute of Materials, Minerals and Mining, *www.instmat.co.uk*
The Minerals, Metals & Materials Society, *www.tms.org/TMSHome.html*
UK Steel, *www.eef.org.uk/uksteel*
World Steel Association, *www.worldsteel.org*

Standards organizations

ASTM, *www.astm.org*
BSI, *www.bsi.org.uk*
DIN Deutsches Institut für Normung e.V, *www.din.de*
International Standards Organisation, *www.iso.org*
National Standards Authority of Ireland, *www.nsai.ie*

Miscellaneous

Hazardous materials (1), *www.phmsa.dot.gov/hazmat*
Hazardous materials (2), *http://ull.chemistry.uakron.edu/erd*
Information for Physical Chemists, *www.liv.ac.uk/Chemistry/Links/links.html*
K&K Associates' thermal connection, *www.tak2000.com*

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Appendix E: Exercises

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E.1 INTRODUCTION TO EXERCISES

These exercises are designed to develop facility in selecting materials, processes, and shape, and in devising hybrid materials when no monolithic material completely meets the design requirements. Each exercise is accompanied by a worked solution. They are organized into the 12 sections listed in the Contents.

The early exercises are easy. Those that follow lead the reader through the use of *material properties* and *simple solutions to mechanics problems*, drawing on data and results contained in Appendices A and B; the use of *material property charts*; techniques for the translation of design requirements to

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identify *constraints* and *objectives*; the derivation of *indices*, *screening*, *ranking*, *and multiobjective optimization*; coupled choice of *material* and *shape*; devising *hybrids*; and the choice of materials to meet *environmental criteria*.

Three important points:

1. Selection problems are open-ended and, generally, underspecified; there is seldom a single, correct answer. The proper answer is sensible translation of the design requirements into material constraints and objectives, applied to give a "short list" of potential candidates with commentary suggesting what supporting information would be needed to narrow the choice further.
2. The positioning of selection lines on charts is a matter of judgment. The goal is to place the lines such that they leave an adequately large "short list" of candidates (aim for four or so), drawn, if possible, from more than one class of material.
3. A request for a selection based on one material index alone (such as $M = E^{1/2}/\rho$) is correctly answered by listing the subset of materials that maximize this index. But a request for a selection of materials for a component—say a wing spar (a light, stiff beam for which the index is $M = E^{1/2}/\rho$)—requires more: Some materials with high $E^{1/2}/\rho$, such as silicon carbide, are unsuitable for obvious reasons. It is a poor answer that ignores common sense and experience and fails to add further constraints to incorporate them. Students should be encouraged to discuss the implications of their selection and to suggest further selection stages.

The best way to use the charts that are a feature of the book is to make clean copies (or download them from *www.grantadesign.com*) on which you can draw, try out alternative selection criteria, write comments, and so forth. Although the book itself is copyrighted, the reader is authorized to make copies of the charts and to reproduce these, with proper reference to their source, as he or she wishes.

All the materials selection problems can be solved using the CES software, which is particularly effective when multiple criteria and unusual indices are involved.

E.2 MATERIAL EVOLUTION IN PRODUCTS (CHAPTER 1)

E2.1 Use Google to research the history and uses of one of the following materials:

- Tin
- Glass
- Cement

- Titanium
- Carbon fiber

Present the result as a short report of about 100 to 200 words (roughly half a page).

E2.2 Research, at the level of the mini-case studies in this chapter, the evolution of material use in

- Writing implements
- Watering cans
- Bicycles
- Small-boat building
- Book binding

E.3 DEVISING CONCEPTS (CHAPTER 2)

These two examples illustrate the way in which concepts are generated. The left part of each diagram describes a physical principle according to which the need might be met; the right part elaborates, suggesting how the principle might be used.

E3.1 Concepts and embodiments for dust removers We met the need for a "device to remove household dust" in Chapter 1, with examples of established solutions. Now it is time for more creative thinking. Devise as many concepts to meet this need as you can. Nothing, at the concept stage, is too far-fetched; decisions about practicality and cost come later, at the detail stage, so think along the lines shown in Figure 2.2 of the main text and list concepts and outline embodiments as block diagrams like those shown in the figure.

E3.2 Cooling power electronics Microchips, particularly those for power electronics, get hot. If they get too hot they cease to function. The need: a scheme for removing heat from power microchips. Devise concepts to meet the need and sketch an embodiment of one of them, laying out your ideas in the way suggested in Exercise E2.1.

E.4 USING MATERIAL PROPERTIES (CHAPTER 3)

These exercises introduce the reader to two useful resources: the data sheets of Appendices A and B.

E4.1 A cantilever beam has a length $L = 50$ mm, a width $b = 5$ mm, and a thickness $t = 1$ mm. It is made of an aluminum alloy. By how much will the end deflect under an end-load of 5 N? Use data from Appendix A.4 for the (mean) value of Young's modulus of aluminum alloys, the equation for the elastic deflection of a cantilever from Appendix B.3, and for the second moment of a beam from Appendix B.2 to find out.

E4.2 A spring, wound from stainless steel wire with a wire diameter $d = 1$ mm, has $n = 20$ turns of radius $R = 10$ mm. How much will it extend when loaded with a mass P of 1 kg? Assume the shear modulus G of stainless steel to be $3/8E$, where E is Young's modulus; retrieve this from Appendix A.4, and use the expression for the extension of springs from Appendix B.6 to find out.

E4.3 A thick-walled tube has an inner radius $r_i = 10$ mm and an outer radius $r_o = 15$ mm. It is made from polycarbonate, PC. What is the maximum torque that the tube can carry without the onset of yield? Retrieve the (mean) yield strength σ_y of PC from Appendix A.5, the expression for the torque at onset of yield from Appendix B.6, and that for the polar moment of a thick-walled tube from Appendix B.2 to find out.

E4.4 A round bar, 20 mm in diameter, has a shallow circumferential notch with a depth $c = 1$ mm and a root radius $r = 10$ microns. The bar is made of a low carbon steel with a yield strength of $\sigma_y = 250$ MPa. It is loaded axially with a nominal stress σ_{nom} (the axial load divided by the unnotched area). At what value of σ_{nom} will yield first commence at the root of the notch? Use the stress concentration estimate of Appendix B.9 to find out.

E4.5 An acrylic (PMMA) window is clamped in a low carbon steel frame at $T = 20°C$. The temperature falls to $T = -20°C$, putting the window under tension because the thermal expansion coefficient of PMMA is larger than that of steel. If the window was stress-free at 20°C, what stress does it carry at −20°C? Use the result that the biaxial stress caused by a biaxial strain difference $\Delta\varepsilon$ is

$$\sigma = \frac{E\Delta\varepsilon}{1-\nu}$$

where E is Young's modulus for PMMA, and Poisson's ratio $\nu = 0.33$. You will find data for expansion coefficients in Table A.7 and for moduli in Table B.5. Use mean values.

E4.6 The PMMA window described in Exercise E4.5 has a contained crack of length $2a = 0.5$ mm. If the maximum tensile stress that is anticipated in the window is $\sigma = 20$ MPa, will the crack propagate? Choose an appropriate equation for crack propagation from Appendix B.10 and data for the fracture toughness K_{1C} of PMMA from Appendix A.6 to calculate the length of crack that is just unstable under this tensile stress.

E4.7 A flywheel with a radius $R = 200$ mm is designed to spin up to 8000 rpm. It is proposed to make it out of cast iron, but the casting shop can guarantee only that it will have no crack-like flaws greater than $2a = 2$ mm in length. Use the expression for the maximum stress in a spinning disk in Appendix B.7, that for the stress intensity at a small enclosed crack from

Appendix B.10, and data for cast iron from Appendices A.3 and A.6 to establish if the flywheel is safe. Take Poisson's ratio ν for cast iron to be 0.33.

E4.8 You wish to assess, approximately, the thermal conductivity λ of polyethylene (PE). To do so you block one end of a PE pipe with a wall thickness of $x = 3$ mm and diameter of 30 mm and fill it with boiling water while clutching the outside with your other hand. You note that the outer surface of the pipe first becomes appreciably hot at time $t \approx 18$ seconds after filling the inside with water. Use this information, plus data for specific heat C_p and density ρ of PE from Appendices A.3 and A.8 to estimate λ for PE. How does your result compare with the listed value in Table A.7?

E4.9 The capacitance C of a condenser with two plates each of area A separated by a dielectric of thickness t is

$$C = \varepsilon_r \varepsilon_o \frac{A}{t}$$

where ε_o is the permittivity of free space and ε_r is the dielectric constant of the material between the plates. Select a dielectric by scanning data in Appendix A.9 (a) to maximize C and (b) to minimize it, for a given A and t.

E4.10 It is proposed to replace the cast iron casing of a power tool with one with precisely the same dimension molded from nylon. Will the material cost of the nylon casing be greater or less than that made of cast iron? Use data from Appendices A.3 and A.11 to find out.

E.5 USING MATERIAL PROPERTY CHARTS (CHAPTER 4)

The 20 exercises in this section involve the simple use of the charts in Chapter 4 to find materials with given property profiles. They are answered by placing selection lines on the appropriate chart and reading off the materials that lie on the appropriate side of the line. It is a good idea to present the results as a table. All can be solved by using the printed charts.

If the CES Edu Materials Selection software is available, these exercises can be solved by its use. This involves first creating the chart, then applying the appropriate box or line selection. The results, at Level 1 or 2, are the same as those read from the hard-copy charts (most of which were made using the Level 1/2 database). The software offers links to processes, allows a wider search by using the Level 3 database, and gives access to supporting information via the "Search Web" function.

E5.1 A component is at present made from brass, a copper alloy. Use the Young's modulus–density (E–ρ) chart shown in Figure 4.3 to suggest three

other metals that, in the same shape, would be stiffer. "Stiffer" means a higher value of Young's modulus.

E5.2 Use the Young's modulus–density (E–ρ) chart shown in Figure 4.3 to identify materials with both a modulus $E > 50$ GPa and a density $\rho < 2000$ kg/m^3.

E5.3 Use the Young's modulus–density (E–ρ) chart shown in Figure 4.3 to find (a) metals that are stiffer and less dense than steels and (b) materials (not just metals) that are both stiffer and less dense than steel.

E5.4 Use the E–ρ chart shown in Figure 4.3 to identify metals with both $E > 100$ GPa and $E/\rho > 0.02$ GPa/(kg/m^3).

E5.5 Use the E–ρ chart shown in Figure 4.3 to identify materials with both $E > 100$ GPa and $E^{1/3}/\rho > 0.003$ (GPa)$^{1/3}$/(kg/m^3). Remember that, on taking logs, the index $M = E^{1/3}/\rho$ becomes

$$Log\,(E) = 3Log\,(\rho) + 3Log\,(M)$$

and that this plots as a line of slope 3 on the chart, passing through the point $E = 27$ when $\rho = 1000$ in the units on the chart.

E5.6 Use the E–ρ chart shown in Figure 4.3 to establish whether woods have a higher specific stiffness E/ρ than epoxies.

E5.7 Do titanium alloys have a higher or lower specific strength (strength/density, σ_f/ρ) than the best steels? This is important when you want strength at low weight (landing gear of aircraft, mountain bikes). Use the σ_f/ρ chart of Figure 4.4 to decide.

E5.8 Use the modulus–strength (E–σ_f) chart shown in Figure 4.5 to find materials that have $E > 10$ GPa and $\sigma_f \geq 1000$ Mpa.

E5.9 Are the fracture toughnesses K_{1C} of the common polymers polycarbonate, ABS, and polystyrene larger or smaller than the engineering ceramic alumina, Al_2O_3? Are their toughnesses $G_{1C} = K_{1C}^2/E$ larger or smaller? The K_{1C}–E chart in Figure 4.7 will help.

E5.10 Use the fracture toughness–modulus chart (Figure 4.7) to find materials that have a fracture toughness K_{1C} greater than 100 MPa.m$^{1/2}$ and a toughness $G_{1C} = K_{1C}^2/E$ (shown as contours in Figure 4.7) greater than 10 kJ/m^3.

E5.11 The elastic deflection at fracture (the "resilience") of an elastic-brittle solid is proportional to the failure strain, $\varepsilon_{fr} = \sigma_{fr}/E$, where σ_{fr} is the stress that will cause a crack to propagate:

$$\sigma_{fr} = \frac{K_{1C}}{\sqrt{\pi c}}$$

Here K_{1C} is the fracture toughness and c is the length of the longest crack the materials may contain. Thus

$$\varepsilon_{fr} = \frac{1}{\sqrt{\pi c}}\left(\frac{K_{1C}}{E}\right)$$

Materials that can deflect elastically without fracturing are therefore those with large values of K_{1C}/E. Use the K_{1C}–E chart shown in Figure 4.7 to identify the class of materials with $K_{1C} > 1$ MPa.m$^{1/2}$ and high values of K_{1C}/E.

E5.12 One criterion for design of a safe pressure vessel is that it should leak before it breaks: The leak can be detected and the pressure released. This is achieved by designing the vessel to tolerate a crack of length equal to the thickness t of the pressure vessel wall, without failing by fast fracture. The safe pressure p is then

$$p \leq \frac{4}{\pi}\frac{1}{R}\left(\frac{K_{1C}^2}{\sigma_f}\right)$$

where σ_f is the elastic limit, K_{1C} is the fracture toughness, and R is the vessel radius. The pressure is maximized by choosing the material with the greatest value of

$$M = \frac{K_{1C}^2}{\sigma_y}$$

Use the $K_{1C} - \sigma_f$ chart shown in Figure 4.8 to identify three alloys that have particularly high values of M.

E5.13 A material is required for the blade of a rotary lawn mower. Cost is a consideration. For safety reasons, the designer specified a minimum fracture toughness for the blade: It is $K_{1C} > 30$ MPa m$^{1/2}$. The other mechanical requirement is for high hardness, H, to minimize blade wear. Hardness, in applications like this one, is related to strength:

$$H \approx 3\sigma_y$$

where σ_f is the strength (Chapter 4 gives a fuller definition). Use the K_{1C}–σ_f chart shown in Figure 4.8 to identify three materials that have $K_{1C} > 30$ MPa m$^{1/2}$ and the highest possible strength. To do this, position a "K_{1C}" selection line at 30 MPa m$^{1/2}$ and then adjust a "strength" selection line such that it just admits three candidates. Use the cost chart shown in Figure 4.19 to rank your selection by material cost, hence making a final selection.

E5.14 Bells ring because they have a low loss (or damping) coefficient, η; a high damping gives a dead sound. Use the loss coefficient–modulus (η–E) chart shown in Figure 4.9 to identify material that should make good bells.

E5.15 Use the loss coefficient–modulus (η–E) chart (Figure 4.9) to find metals with the highest possible damping.

E5.16 Use the thermal conductivity–electrical resistivity (λ–ρ_e) chart (Figure 4.10) to find three materials with high thermal conductivity, λ, and high electrical resistivity ρ_e.

E5.17 The window through which the beam emerges from a high-powered laser must obviously be transparent to light. Even then, some of the energy of the beam is absorbed in the window and can cause it to heat and crack. This problem is minimized by choosing a window material with a high thermal conductivity λ (to conduct the heat away) and a low expansion coefficient α (to reduce thermal strains); that is, by seeking a window material with a high value of

$$M = \lambda/\alpha$$

Use the α–λ chart shown in Figure 4.12 to identify the best material for an ultra-high-powered laser window.

E5.18 Use the maximum service temperature (T_{max}) chart (Figure 4.14) to find polymers that can be used above 200°C.

E5.19

a. Use the Young's modulus–relative cost (E–$C_{v,R}$) chart (Figure 4.18) to find the cheapest materials with a modulus, E, greater than 100 GPa.
b. Use the strength–relative cost (σ_f–$C_{v,R}$) chart (Figure 4.19) to find the cheapest materials with a strength, σ_f, above 100 MPa.

E5.20 Use the friction coefficient chart (Figure 4.15) to find two materials with exceptionally low coefficients of friction.

E.6 TRANSLATION: CONSTRAINTS AND OBJECTIVES (CHAPTERS 5 AND 6)

Translation is the task of re-expressing design requirements in terms that enable material and process selection. Tackle the exercises by formulating the answers to the questions in this table. Don't try to model the behavior at this point (that comes in later exercises). Just think out what the component does, and list the constraints that this imposes on material choice, including processing requirements.

Function	What does the component do?
Constraint	Which essential conditions must be met?
Objective	What is to be maximized or minimized?
Free variable	Which parameters of the problem is the designer free to change?

Here it is important to recognize the distinction between constraints and objectives. As the table says, a constraint is an essential condition that must be met, usually expressed as a **limit** on a material or process attribute. An objective is a quantity for which an **extremum** (a maximum or minimum) is sought, frequently cost, mass or volume, but there are others, several of which appear in the exercises that follow. Take the example of a bicycle frame. It must have a certain stiffness and strength. If it is not stiff and strong enough it will not work, but it is never required to have infinite stiffness or strength. Stiffness and strength are therefore constraints that become limits on modulus, elastic limit, and shape. If the bicycle is for sprint racing, it should be as light as possible—if you could make it infinitely light, that would be best of all. Minimizing mass, here, is the objective, perhaps with an upper limit (a constraint) on cost. If instead it is a shopping bike to be sold through supermarkets, it should be as cheap as possible—the less expensive it is, the more will be sold. This time minimizing cost is the objective, possibly with an upper limit (a constraint) on mass. For most bikes, of course, minimizing mass and cost are both objectives, and then trade-off methods are needed. They come later. For now use judgment to choose the single most important objective and make all others into constraints.

Here are two rules-of-thumb, useful in many "translation" exercises. Many applications require sufficient fracture toughness for the component to survive mishandling and accidental impact during service; a totally brittle material (like untoughened glass) is unsuitable. Then a necessary constraint is that of "adequate toughness." This is achieved by requiring that the fracture toughness K_{1C}15 MPa. $m^{1/2}$. Other applications require some ductility, sufficient to allow stress redistribution under loading points, and some ability to bend or shape the material plastically. This is achieved by requiring that the (tensile) ductility ε_f be 2%.

(If the CES software is available, it can be used to impose the constraints and to rank the survivors using the objective.)

E6.1 A material is required for the windings of an electric air furnace capable of temperatures up to 1000°C. Think out what attributes a material must have if it is to be made into windings and function properly in a furnace. List the function and the constraints; set the objective to "minimize cost" and the free variables to "choice of material."

E6.2 A material is required to manufacture office scissors. Paper is an abrasive material, and scissors sometimes encounter hard obstacles like staples. List the function and the constraints; set the objective to "minimize cost" and the free variable to "choice of material."

E6.3 A material is required for a heat exchanger to extract heat from geothermally heated, saline water at 120°C (and thus under pressure). List the function and the constraints; set the objective to "minimize cost" and the free variable to "choice of material."

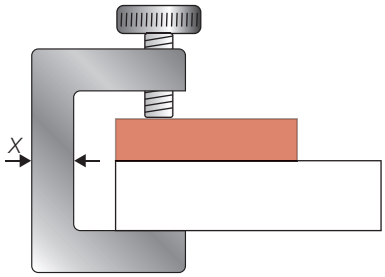


FIGURE E.1

E6.4 C-clamp is required for processing electronic components at temperatures up to 450°C (Figure E.1). It is essential that the clamp have as low a thermal inertia as possible so that it reaches temperature quickly, and it must not charge up when exposed to an electron beam. The time t it takes a component of thickness x to reach thermal equilibrium when the temperature is suddenly changed (which is a transient heat flow problem) is

$$t \approx \frac{x^2}{2a}$$

where the thermal diffusivity $a = \lambda/\rho C_p$ and λ is the thermal conductivity, ρ the density, and C_p the specific heat.

List function, constraints, and objective; set the free variables to "choice of material."

E6.5 A furnace is required to sinter powder-metal parts. It operates continuously at 650°C while the parts are fed through on a moving belt. You are asked to select a material for furnace insulation to minimize heat loss and thus make the furnace as energy efficient as possible. For reasons of space the insulation is limited to a maximum thickness of $x = 0.2$ m. List the function, constraints, objective, and free variable.

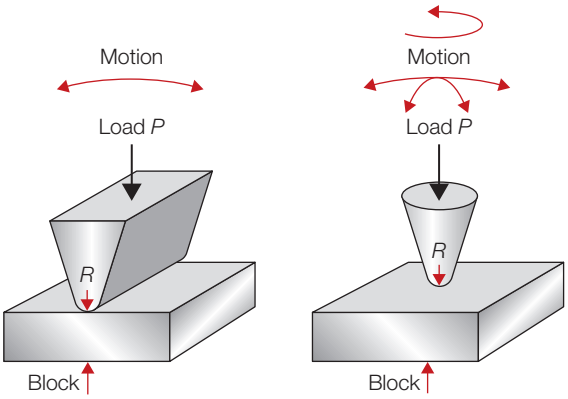


FIGURE E.2

E6.6 Ultraprecise bearings that allow a rocking motion make use of knife-edges or pivots (Figure E.2). As the bearing rocks, it rolls, translating sideways by a distance that depends on the radius of contact. The further it rolls, the less precise is its positioning, so the smaller the radius of contact R the better. But the smaller the radius of contact, the greater is the contact pressure (F/A). If this exceeds the hardness H of either face of the bearing, it will be damaged. Elastic deformation is bad too: It flattens the contact, increasing the contact area and the roll.

A rocking bearing is required to operate in a micro-chip fabrication unit using fluorine gas at 100°C, followed by e-beam processing requiring that all structural parts of the equipment can be

earthed to prevent stray charges. Translate the requirements into material selection criteria, listing function, constraints, objective, and free variable.

E6.7 The standard CD ("jewel" case) cracks easily and, if broken, can scratch the CD. Jewel cases are made of injection-molded polystyrene, chosen because it is transparent, cheap, and easy to mold. A material is sought to make CD cases that do not crack so easily. The case must still be transparent, able to be injection-molded, and able to compete with polystyrene in cost.

E6.8 A storage heater captures heat over a period of time, then releases it, usually to an air stream, when required. Those for domestic heating store solar energy or energy from cheap off-peak electricity and release it slowly during the cold part of the day. Those for research release the heat to a supersonic air stream to test system behavior in supersonic flight. What is a good material for the core of a compact storage material capable of temperatures up to 120°C?

E.7 DERIVING AND USING MATERIAL INDICES (CHAPTERS 5 AND 6)

The exercises in this section give practice in deriving indices.

a. Start each by listing function, constraints, objectives, and free variables; without having those straight, you will get in a mess. Then write down an equation for the objective. Consider whether it contains a free variable other than material choice; if it does, identify the constraint that limits it, substitute, and read off the material index.

b. If the CES Edu software is available, use it to apply the constraints and rank the survivors using the index (start with the Level 2 database). Are the results sensible? If not, what constraint has been overlooked or incorrectly formulated?

E7.1 Aperture grills for cathode ray tubes There are two types of cathode ray tube (CRT). In the older technology, color separation is achieved by using a *shadow mask*: a thin metal plate with a grid of holes that allow only the correct beam to strike a red, green, or blue phosphor. A shadow mask can heat up and distort at high brightness levels ("doming"), causing the beams to miss their target, and giving a blotchy image.

To avoid this, shadow masks are made of Invar, a nickel alloy with a near-zero expansion coefficient between room temperature and 150°C. It is a consequence of shadow-mask technology that the glass screen of the CRT curves inward on all four edges, increasing the probability of reflected glare.

Sony's "Trinitron" technology overcame this problem and allowed greater brightness by replacing the shadow mask with an *aperture grill* of fine vertical wires, each about 200 μm in thickness, that allows the intended beam to

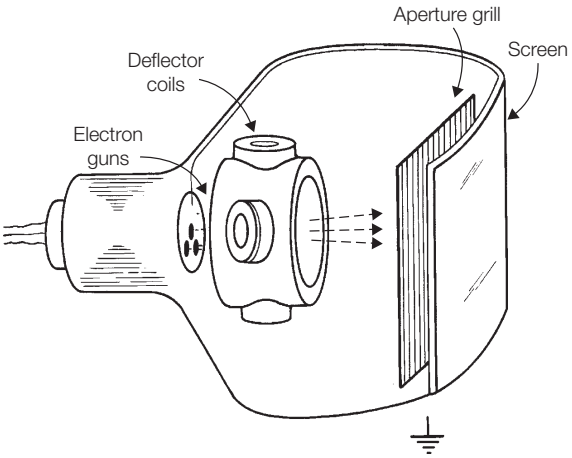


FIGURE E.3

strike either the red, the green, or the blue phosphor to create the image (see Figure E.3). The glass face of the Trinitron tube was curved in one plane only, reducing glare.

The wires of the aperture grill are tightly stretched, so that they remain taut even when hot—it is this tension that allows the greater brightness. What index guides the choice of material to make them? The table summarizes the requirements.

Function	Aperture grill for CRT
Constraints	Wire thickness and spacing specified Must carry pre-tension without failure Electrically conducting to prevent charging Able to be drawn to wire
Objective	Maximize permitted temperature rise without loss of tension
Free variable	Choice of material

E7.2 Material indices for elastic beams with differing constraints (Figure E.4) Start each of the four parts of this problem by listing the function, the objective, and the constraints. You will need the equations for the deflection of a cantilever beam with a square cross-section $t \times t$, given in Appendix B.3. The two that matter are that for the deflection δ of a beam of length L under an end load F:

$$\delta = \frac{FL^3}{3EI}$$

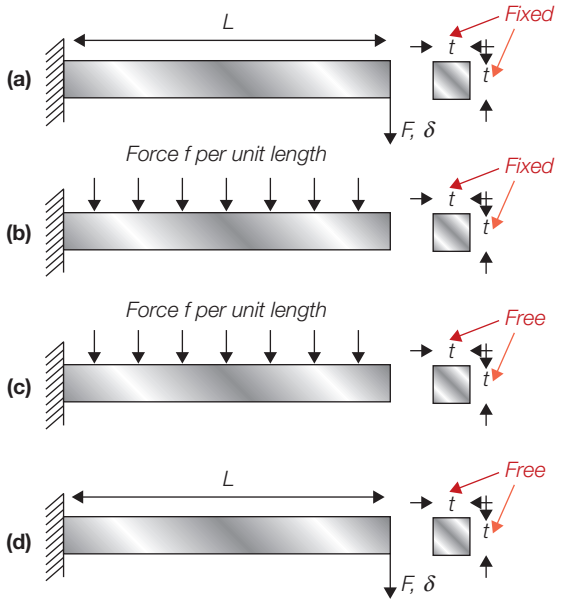


FIGURE E.4

and that for the deflection of a beam under a distributed load f per unit length:

$$\delta = \frac{1}{8}\frac{fL^4}{EI}$$

where $I = t^4/12$. For a self-loaded beam $f = \rho Ag$, where ρ is the density of the material of the beam, A its cross-sectional area, and g the acceleration due to gravity.

a. Show that the best material for a cantilever beam of given length L and *given* (i.e., fixed) square cross-section ($t \times t$) that will deflect least under a given end load F is that with the largest value of the index $M = E$, where E is Young's modulus (neglect self-weight). (See Figure E.4(a).)

b. Show that the best material choice for a cantilever beam of given a length L and with a given section ($t \times t$) that will deflect least under its own self-weight is that with the largest value of $M = E/\rho$, where ρ is the density. (Figure E.4(b).)

c. Show that the material index for the lightest cantilever beam of length L and square section (not given, i.e., the area is a free variable) that will not deflect by more than δ under its own weight is $M = E/\rho^2$. (See Figure E.4(c).)

d. Show that the lightest cantilever beam of length L and square section (area free) that will not deflect by more than δ under an end load F is that made of the material with the largest value of $M = E^{1/2}/\rho$ (neglect self weight). (See Figure E.4(d).)

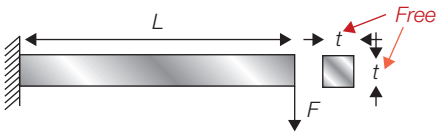


FIGURE E.5

E7.3 Material index for a light, strong beam In stiffness-limited applications, it is elastic deflection that is the active constraint: It limits performance (Figure E.5). In strength-limited applications, deflection is acceptable provided the component does not fail; strength is the active constraint. Derive the material index for selecting materials for a beam of length L, specified strength, and minimum weight. For simplicity, assume the beam to have a solid square cross-section $t \times t$. You will need the equation for the failure load of a beam (Appendix B, Section B.4). It is

$$F_f = \frac{I\sigma_f}{y_m L}$$

where y_m is the distance between the neutral axis of the beam and its outer filament and $I = t^4/12 = A^2/12$ is the second moment of the cross-section. The table itemizes the design requirements.

Function	Beam
Constraints	Length L is specified Beam must support a bending load F without yield or fracture
Objective	Minimize the mass of the beam
Free variables	Cross-section area, A Choice of material

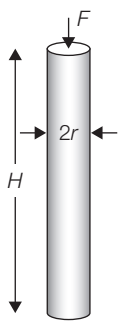


FIGURE E.6

E7.4 Material index for a cheap, stiff column (Figure E.6) In the preceding two exercises the objective has been that of minimizing weight. There are many others. In the selection of a material for a spring, the objective is that of maximizing the elastic energy it can store. In seeking materials for thermal-efficient insulation for a furnace, the best are those with the lowest thermal conductivity and heat capacity. And most common of all is the wish to minimize cost. So here is an example involving cost.

Columns support compressive loads: the legs of a table; the pillars of the Parthenon. Derive the index for selecting materials for the cheapest cylindrical

column of specified height, H, that will safely support a load F without buckling elastically. You will need the equation for the load F_{crit} at which a slender column buckles. It is

$$F_{crit} = \frac{n^2 \pi^2 E I}{H^2}$$

where n is a constant that depends on the end constraints and $I = \pi r^4/4 = A^2/4\pi$ is the second moment of area of the column (see Appendix B for both). The following table lists the requirements.

Function	Cylindrical column
Constraints	Length L specified Column must support a compressive load F without buckling
Objective	Minimize the material cost of the column
Free variables	Cross-section area A Choice of material

E7.5 Indices for stiff plates and shells Aircraft and space structures make use of plates and shells (Figure E.7). The index depends on the configuration. Here you are asked to derive the material index for

a. a circular plate of radius a carrying a central load W with a prescribed stiffness $S = W/\delta$ and of minimum mass

b. a hemispherical shell of radius a carrying a central load W with a prescribed stiffness $S = W/\delta$ and of minimum mass, as shown in the figures

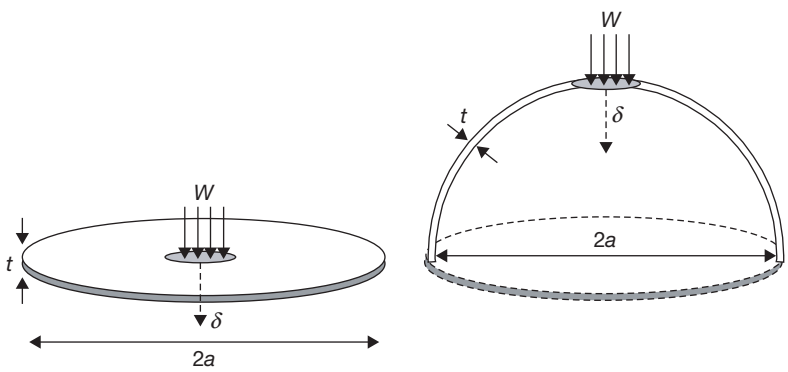


FIGURE E.7

Use the two results listed below for the mid-point deflection δ of a plate or spherical shell under a load W applied over a small central, circular area.

$$\text{Circular plate: } \delta = \frac{3}{4\pi}\frac{Wa^2}{Et^3}(1-\nu^2)\left(\frac{3+\nu}{1+\nu}\right)$$

$$\text{Hemispherical shell: } \delta = A\frac{Wa}{Et^2}(1-\nu^2)$$

in which $A \approx 0.35$ is a constant. Here E is Young's modulus, t is the thickness of the plate or shell, and ν is Poisson's ratio. Poisson's ratio is almost the same for all structural materials and can be treated as a constant. The table summarizes the requirements.

Function	Stiff circular plate or stiff hemispherical shell
Constraints	Stiffness S under central load W specified Radius a of plate or shell specified
Objective	Minimize the mass of the plate or shell
Free variables	Plate or shell thickness t Choice of material

E7.6 The C-clamp in more detail (Figure E.8) Exercise E4.4 introduced the C-clamp for processing of electronic components. The clamp has a square cross-section of width x and given depth b. It is essential that the clamp have low thermal inertia so that it reaches temperature quickly. The time t it takes a component of thickness x to reach thermal equilibrium when the temperature is suddenly changed (a transient heat flow problem) is

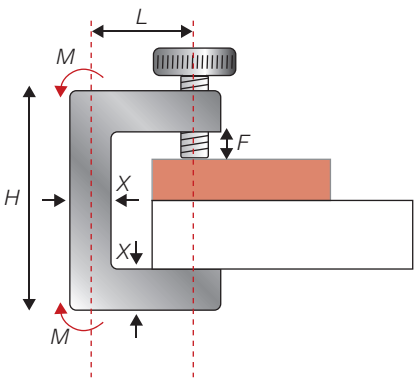


FIGURE E.8

$$t \approx \frac{x^2}{2a}$$

where the thermal diffusivity $a = \lambda/\rho C_p$ and λ is the thermal conductivity, ρ the density, and C_p the specific heat.

The time to reach thermal equilibrium is reduced by making the section x thinner, but it must not be so thin that it fails in service. Use this constraint to eliminate x in the equation above, thereby deriving a material index for the clamp. Use the fact that the clamping

force F creates a moment on the body of the clamp of $M = FL$, and that the peak stress in the body is given by

$$\sigma = \frac{x}{2}\frac{M}{I}$$

where $I = bx^3/12$ is the second moment of area of the body. The table summarizes the requirements.

Function	C-clamp of low thermal inertia
Constraints	Depth b specified Must carry clamping load F without failure
Objective	Minimize time to reach thermal equilibrium
Free variables	Width of clamp body x Choice of material

E7.7 Springs for trucks For the design of vehicle suspension it is desirable to minimize the mass of all components. You have been asked to select a material and dimensions for a light spring to replace the steel leaf spring of an existing truck suspension. The existing leaf-spring is a beam, shown schematically in the Figure E.9. The new spring must have the same length L and stiffness S as the existing one, and must deflect through a maximum safe displacement δ_{max} without failure. The width b and thickness t are free variables.

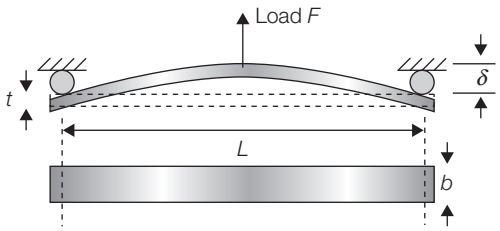


FIGURE E.9

Derive a material index for the selection of a material for this application. Note that this is a problem with two free variables: b and t; and there are two constraints, one on safe deflection δ_{max} and the other on stiffness S. Use the two constraints to fix free variables. The following table catalogs the requirements.

Function	Leaf spring for truck
Constraints	Length L specified Stiffness S specified Maximum displacement δ_{max} specified
Objective	Minimize mass
Free variables	Spring thickness t Spring width b Choice of material

You will need the equation for the mid-point deflection of an elastic beam of length L loaded in three-point bending by a central load F:

$$\delta = \frac{1}{48}\frac{FL^3}{EI}$$

and that for the deflection at which failure occurs:

$$\delta_{\max} = \frac{1}{6}\frac{\sigma_f L^2}{tE}$$

where I is the second moment of area; for a beam of rectangular section, $I = bt^3/12$ and E and σ_f are the modulus and failure stress of the material of the beam. (See Appendix B.)

E7.8 Fin for a rocket A tube-launched rocket has stabilizing fins at its rear (Figure E.10). During launch the fins experience hot gas at $T_g = 1700°$ C for a time $t = 0.3$ seconds. It is important that the fins survive launch without surface melting. Suggest a material index for selecting a material for the fins. The following table summarizes the requirements.

Function	High-heat-transfer rocket fins
Constraints	All dimensions specified Must not suffer surface melting during exposure to gas at 1700°C for 0.3 sec
Objective	Minimize the surface temperature rise during firing Maximize the melting point of the material
Free variable	Choice of material

This is tricky. Heat enters the surface of the fin by transfer from the gas. If the heat transfer coefficient is h, the heat flux per unit area is

$$q = h(T_g - T_s)$$

where T_s is the surface temperature of the fin—the critical quantity we wish to minimize. Heat diffuses into the fin surface by thermal conduction.

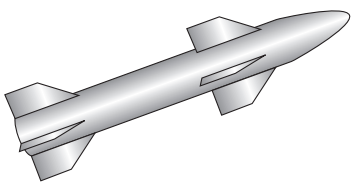


FIGURE E.10

If the heating time is small compared with the characteristic time for heat to diffuse through the fin, a quasi-steady state exists in which the surface temperature adjusts itself such that the heat entering from the gas is equal to that diffusing inward by conduction. This second is equal to

$$q = \lambda \frac{(T_s - T_i)}{x}$$

where λ is the thermal conductivity, T_i is the temperature of the (cold) interior of the fin, and x is a characteristic heat-diffusion length. When the heating time is short (as here) the thermal front, after a time t, has penetrated a distance

$$x \approx (2at)^{1/2}$$

where $a = \lambda/\rho C_p$ is the thermal diffusivity. Substituting this value of x in the previous equation gives

$$q = (\lambda \rho C_p)^{1/2} \frac{(T_s - T_i)}{\sqrt{2t}}$$

where ρ is the density and C_p the specific heat of the material of the fin.

Proceed by equating the two equations for q, solving for the surface temperature T_s to give the objective function. Read off the combination of properties that minimize T_s; it is the index for the problem.

The selection is made by seeking materials with large values of the index and with a high melting point, T_m. If the CES software is available, make a chart with these two as axes and identify materials with high values of the index that also have high melting points.

E.8 MULTIPLE CONSTRAINTS AND OBJECTIVES (CHAPTERS 7 AND 8)

Over constrained problems are normal in materials selection. Often it is just a case of applying each constraint in turn, retaining only those solutions that meet them all. But when constraints are used to eliminate free variables in an objective function (as discussed in Section 9.5 of the text), the "active constraint" method must be used. The first three exercises in this section illustrate problems with multiple constraints.

The remaining two concern multiple objectives and trade-off methods. When a problem has two objectives—minimizing both mass m and cost C of a component, for instance—a conflict arises: The cheapest solution is not the lightest and vice versa. The best combination is sought by constructing a trade-off plot using mass as one axis and cost as the other. The lower envelope of the points on this plot defines the trade-off surface. The solutions that offer the best compromise lie on this surface. To get further we need a penalty function. Define the penalty function

$$Z = C + \alpha m$$

where α is an exchange constant describing the penalty associated with unit increase in mass, or, equivalently, the value associated with a unit decrease. The best solutions are found where the line defined by this equation is tangential to the trade-off surface. (Remember that objectives must be expressed in a form such that a minimum is sought; then a low value of Z, not a high one, is desirable).

When a substitute is sought for an existing material it is better to work with ratios. Then the penalty function becomes

$$Z^* = \frac{C}{C_o} + \alpha^* \frac{m}{m_o}$$

in which the subscript o means properties of the existing material and the asterisk on Z^* and α^* is a reminder that both are now dimensionless. The relative exchange constant α^* measures the fractional gain in value for a given fractional gain in performance.

E8.1 Multiple constraints: A light, stiff, strong tie (Figure E.11) A tie of length L loaded in tension is to support a load F, at minimum weight without failing (implying a constraint on strength) or extending elastically by more than δ (implying a constraint on stiffness, F/δ). The table summarizes the requirements.

a. Follow the method of Chapter 7 to establish two performance equations for the mass, one for each constraint, from which two material indices and one coupling equation linking them are derived. Show that the two indices are

$$M_1 = \frac{\rho}{E} \quad \text{and} \quad M_2 = \frac{\rho}{\sigma_y}$$

and that a minimum is sought for both.

b. Use these and the material chart shown in Figure E.12, which has the indices as axes, to identify candidate materials for the tie (1) when $\delta/L = 10^{-3}$ and (2) when $\delta/L = 10^{-2}$.

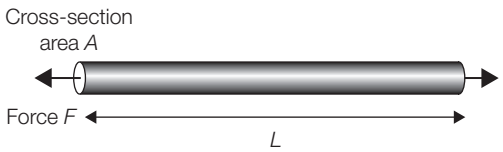


FIGURE E.11

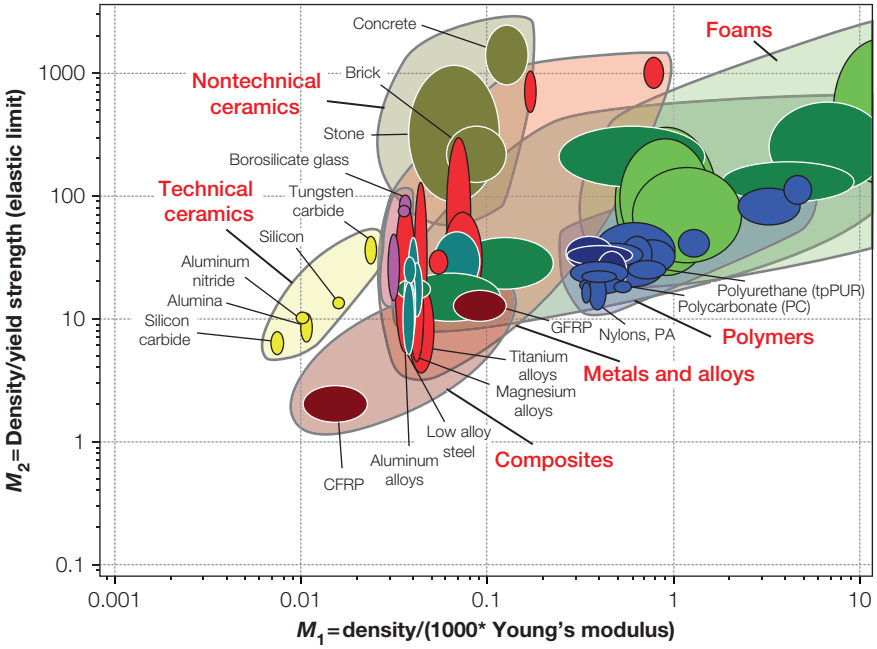


FIGURE E.12

E8.2 Multiple constraints: A light, safe pressure vessel (Figure E.13) When a pressure vessel has to be mobile, its weight becomes important. Aircraft bodies, rocket casings and liquid natural gas containers are examples; they must be light, and at the same time they must be safe; that means that they must not fail by yielding or by fast fracture. What are the best materials for their construction? The following table summarizes the requirements.

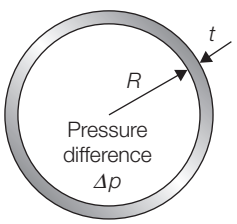


FIGURE E.13

Function	Pressure vessel
Constraints	Must not fail by yielding Must not fail by fast fracture Diameter $2R$ and pressure difference Δp specified
Objective	Minimize mass m
Free variables	Wall thickness t Choice of material

a. First write a performance equation for the mass m of the pressure vessel. Assume, for simplicity, that it is spherical, of specified radius R, and that

the wall thickness t (the free variable) is small compared with R. Then the tensile stress in the wall is

$$\sigma = \frac{\Delta p\, R}{2t}$$

where Δp, the pressure difference across this wall, is fixed by the design. The first constraint is that the vessel should not yield—that is, that the tensile stress in the wall should not exceed σ_y. The second is that it should not fail by fast fracture; this requires that the wall stress be less than $K_{1C}\sqrt{\pi c}$, where K_{1C} is the fracture toughness of the material of which the pressure vessel is made and c is the length of the longest crack that the wall might contain. Use each of these in turn to eliminate t in the equation for m; use the results to identify two material indices:

$$M_1 = \frac{\rho}{\sigma_y} \quad \text{and} \quad M_2 = \frac{\rho}{K_{1C}}$$

and a coupling relation between them. It contains the crack length, c.

b. Figure E.14 shows the chart you will need with the two material indices as axes. Plot the coupling equation onto this figure for two values of c: one of 5 mm, the other of 5 μm. Identify the lightest candidate materials for the vessel for each case.

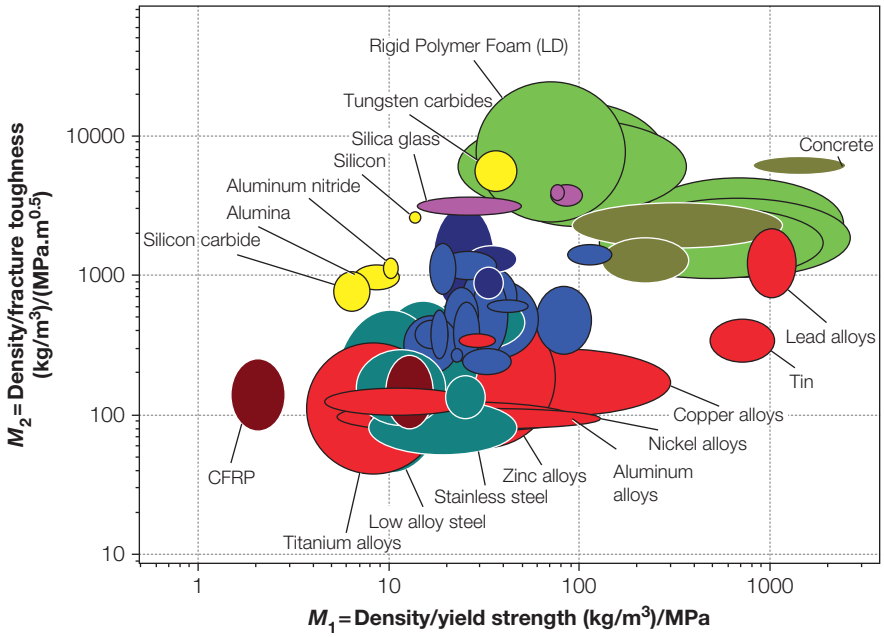


FIGURE E.14

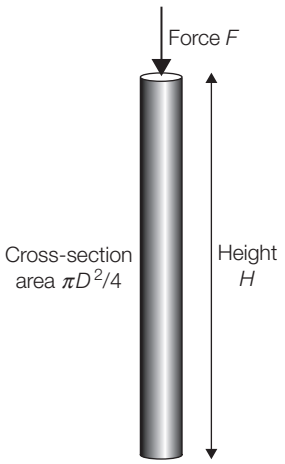


FIGURE E.15

E8.3 A cheap column that must not buckle or crush (Figure E.15) The best choice of material for a light, strong column depends on its aspect ratio: the ratio of its height H to its diameter D. This is because short, fat columns fail by crushing; tall, slender columns buckle instead.

Derive two performance equations for the material cost of a column of a solid circular section and of a specified height H, designed to support a load F (i.e., large compared to its self-load), one using the constraints that the column must not crush, the other that it must not buckle. The following table summarizes the needs.

Function	Column
Constraints	Must not fail by compressive crushing Must not buckle Height H and compressive load F specified
Objective	Minimize material cost C
Free variables	Diameter D Choice of material

a. Proceed as follows

1. Write down an expression for the material cost of the column—its mass times its cost per unit mass C_m.
2. Express the two constraints as equations, and use them to substitute for the free variable to find the cost of the column that will just support the load without failing by either mechanism.
3. Identify the material indices M_1 and M_2 that enter the two equations for the mass, showing that they are

$$M_1 = \left(\frac{C_m\rho}{\sigma_c}\right) \text{ and } M_2 = \left[\frac{C_m\rho}{E^{1/2}}\right]$$

where C_m is the material cost per kg, ρ the material density, σ_c its crushing strength, and E its modulus.

b. Data for six possible candidates for the column are listed in the table that follows. Use these to identify candidate materials when $F = 10^5$ N and $H = 3$ m. Ceramics are admissible here because they have high strength in compression.

Data for Candidate Materials for the Column				
Material	**Density ρ (kg/m^3)**	**Cost/kg C_m ($/kg)**	**Modulus E (MPa)**	**Compression Strength σ_c (MPa)**
Wood (spruce)	700	0.5	10,000	25
Brick	2100	0.35	22,000	95
Granite	2600	0.6	20,000	150
Poured concrete	2300	0.08	20,000	13
Cast iron	7150	0.25	130,000	200
Structural steel	7850	0.4	210,000	300
Al alloy 6061	2700	1.2	69,000	150

c. Figure E.16 shows a material chart with the two indices as axes. Identify and plot coupling lines for selecting materials for a column with $F = 10^5\ N$ and $H = 3\ m$ (the same conditions as previously noted), and for a second column with $F = 10^3\ N$ and $H = 20\ m$.

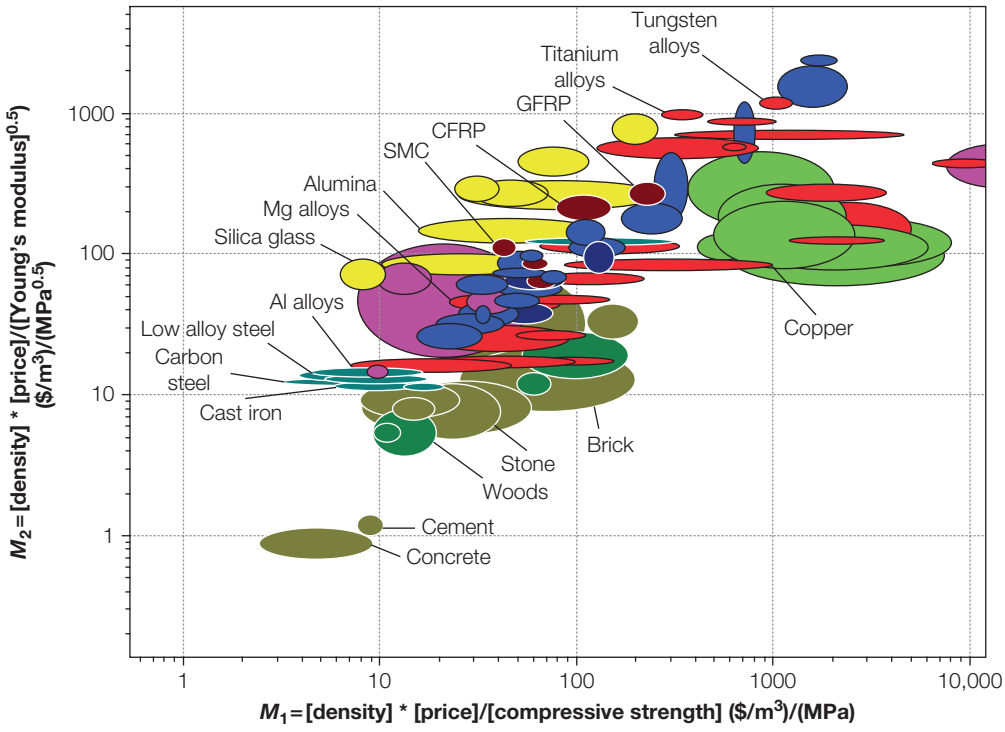


FIGURE E.16

E8.4 A truck's air cylinder Trucks rely on compressed air for braking and other power-actuated systems. The air is stored in one or a cluster of cylindrical pressure tanks like that shown in Figure E.17 (length L, diameter $2R$, hemispherical ends). Most are made of low carbon steel and are heavy. The task: to explore the potential of alternative materials for lighter air tanks, recognizing that there must be a trade-off between mass and cost—if it is too expensive, the truck owner will not want it even if it *is* lighter. The following table summarizes the design requirements.

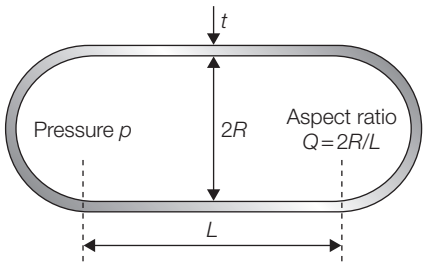


FIGURE E.17

Function	Air cylinder for truck
Constraints	Must not fail by yielding Diameter $2R$ and length L specified, so the ratio $Q = 2R/L$ is fixed
Objectives	Minimize mass m Minimize material cost C
Free variables	Wall thickness t Choice of material

a. Show that the mass and material cost of the tank relative to one made of low carbon steel are given by

$$\frac{m}{m_o} = \left(\frac{\rho}{\sigma_y}\right)\left(\frac{\sigma_{y,o}}{\rho_o}\right) \text{ and } \frac{C}{C_o} = \left(\frac{C_m\rho}{\sigma_y}\right)\left(\frac{\sigma_{y,o}}{C_{m,o}\rho_o}\right)$$

where ρ is the density, σ_y the yield strength, C_m the cost per kg of the material, and the subscript o indicates values for mild steel.

b. Explore the trade-off between relative cost and relative mass, considering the replacement of a mild steel tank with one made, first, of low alloy steel and, second, one made of filament-wound CFRP, using the material properties in the table below. Define a relative penalty function:

$$Z^* = \alpha^* \frac{m}{m_o} + \frac{C}{C_o}$$

where α^* is a relative exchange constant, and evaluate Z^* for $\alpha^* = 1$ and for $\alpha^* = 100$.

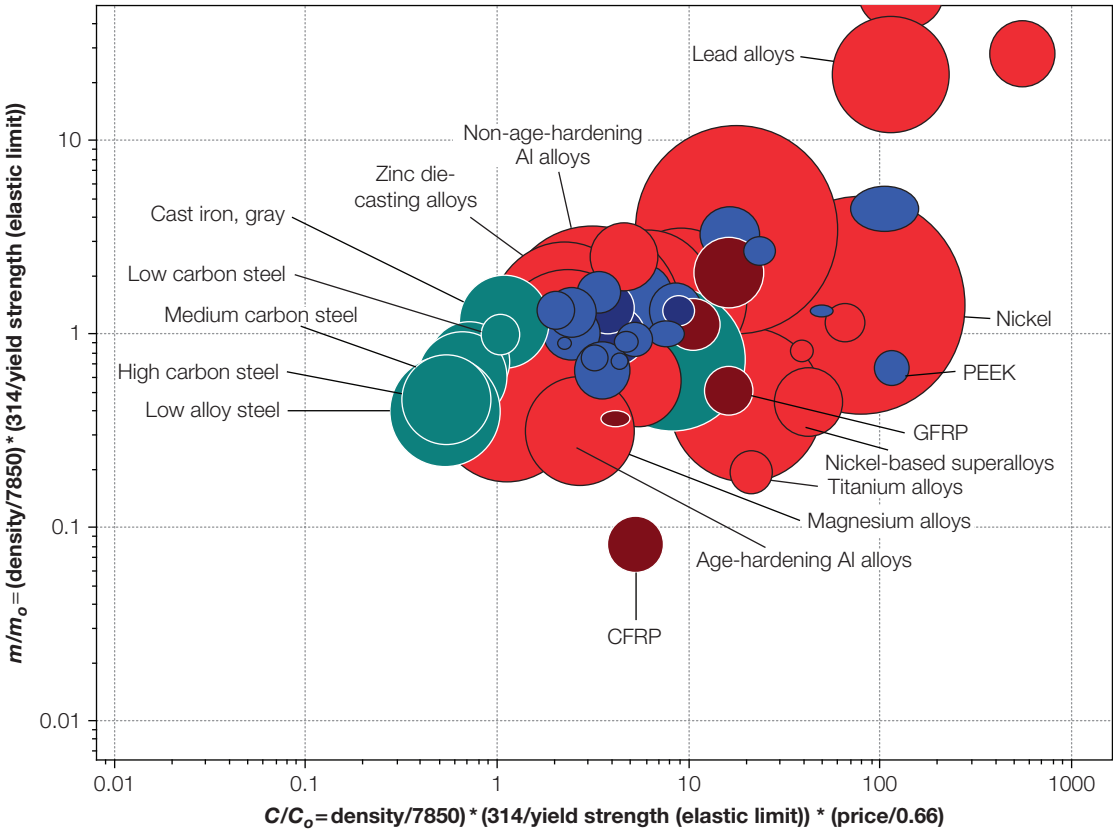


FIGURE E.18

Material	Density ρ (kg/m^3)	Yield Strength σ_c (MPa)	Price per kg C_m ($/kg)
Mild steel	7850	314	0.66
Low alloy steel	7850	775	0.85
CFRP	1550	760	42.1

c. Figure E.18, and the chart below it, shows the axes of m/m_o and C/C_o. Mild steel (here labeled "Low carbon steel") lies at the coordinates (1,1).

Sketch a trade-off surface and plot contours of Z^* that are approximately tangent to the trade-off surface for $\alpha^* = 1$ and for $\alpha^* = 100$. What selections do these suggest?

E8.5 Insulating walls for freezers Freezers and refrigerated trucks (see Figure E.19)have panel walls that provide thermal insulation, and at the



FIGURE E.19

same time are stiff, strong, and light (stiffness to suppress vibration, strength to tolerate rough usage). To achieve this the panels are usually of sandwich construction, with two skins of steel, aluminum, or GFRP (providing the strength) separated by, and bonded to, a low-density insulating core. In choosing the core we seek to minimize thermal conductivity λ and at the same time maximize stiffness, because this allows thinner steel faces and thus a lighter panel while still maintaining the overall panel stiffness. The following table summarizes the design requirements.

Function	Foam for panel-wall insulation
Constraint	Panel-wall thickness specified
Objectives	Minimize foam thermal conductivity λ Maximize foam stiffness, meaning Young's modulus E
Free variable	Choice of material

Figure E.20 shows the thermal conductivity λ of foams plotted against their elastic compliance $1/E$ (the reciprocal of their Young's moduli E, since we

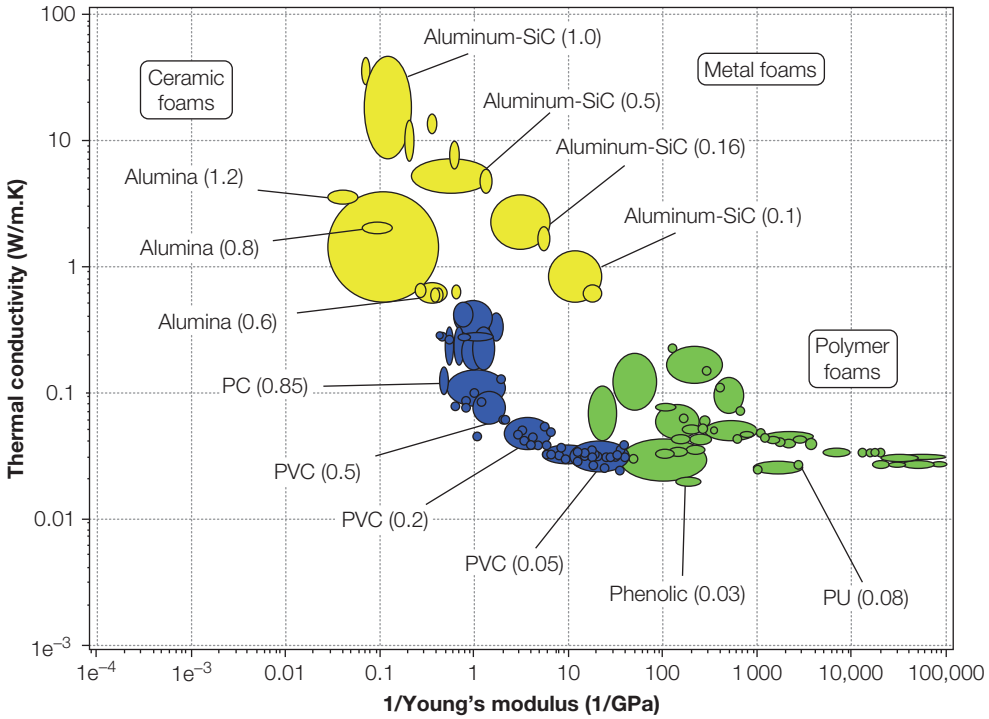


FIGURE E.20

must express the objectives in a form that requires minimization). The numbers in brackets are the densities of the foams in Mg/m^3. The foams with the lowest thermal conductivity are the least stiff; the stiffest have the highest conductivity. Explain the reasoning you would use to select a foam for the truck panel using a penalty function.

E.9 SELECTING MATERIAL AND SHAPE (CHAPTERS 9 AND 10)

The examples in this section relate to the analysis of material and shape from Chapters 9 and 10. They cover the derivation of shape factors and of indices that combine material and shape, and the use of the 4-quadrant chart arrays to explore material and shape combinations. For this last purpose it is useful to have clean copies of the chart arrays shown in Figures 9.9 and 9.12. Like the material property charts, they can be copied from the text without the restrictions of copyright.

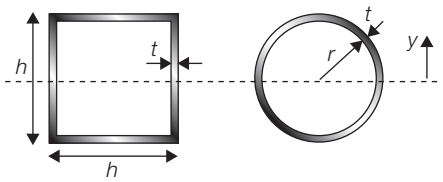


FIGURE E.21

E9.1 Shape factors for tubes (Figure E.21)

a. Evaluate the shape factor φ_B^e for stiffness-limited design in bending of a square box section of outer edge length $h = 100$ mm and wall thickness $t = 3$ mm. Is this shape more efficient than one made of the same material in the form of a tube of diameter $2r = 100$ mm and wall thickness $t = 3.82$ mm (giving it the same mass per unit length m/L)? Treat both as thin-walled shapes.

b. Make the same comparison for the shape factor φ_B^f for strength-limited design.

Use the expressions given in Table 9.3 of the text for the shape factors φ_B^e and φ_B^f.

E9.2 Deriving shape factors for stiffness-limited design

Derive the expression for the shape-efficiency factor for φ_B^e

a. A stiffness-limited design for a beam loaded in bending with each of the three sections shown in Figure E.22 do not assume that the thin-wall approximations is valid

b. A closed circular tube of outer radius $5t$ and wall thickness t

c. a channel section of thickness t, overall flange width $5t$ and overall depth $10t$, bent about its major axis

d. a box section of wall thickness t, and height and width $h_1 = 10t$

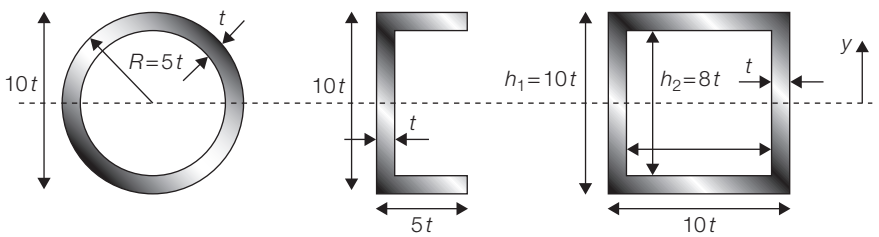


FIGURE E.22

E9.3 Deriving shape factors for strength-limited design

a., b., and c. Determine the shape-efficiency factor φ_B^f for strength limited design in bending, for the same three sections shown in Figure E.22. You will need the results for I for the sections derived in Exercise E9.2.

d. A beam of length L, loaded in bending, must support a specified bending moment M without failing and be as light as possible. Show that to minimize the mass of the beam per unit length m/L, one should select a material and a section shape to maximize the quantity:

$$M = \frac{(\varphi_B^f \sigma_f)^{2/3}}{\rho}$$

where σ_f is the failure stress and ρ the density of the material of the beam; φ_B^f is the shape-efficiency factor for failure in bending.

E9.4 Determining shape factors from stiffness data The elastic shape factor measures the gain in stiffness by shaping, relative to a solid square section of the same area. Shape factors can be determined by experiment. Equation (9.19) in the text gives the mass m of a beam of length L and prescribed bending stiffness S_B with a section of efficiency φ_B^e as

$$m = \left(\frac{12 S_B}{C_1}\right)^{1/2} L^{5/2} \left[\frac{\rho}{(\varphi_B^e E)^{1/2}}\right]$$

where C_1 is a constant that depends on the distribution of load on the beam. Inverting the equation gives

$$\varphi_B^e = \left(\frac{12 L^5 S_B}{C_1 m^2}\right)\left(\frac{\rho^2}{E}\right)$$

Thus if the bending stiffness S_B, mass m, and length L are measured, and the modulus E and density ρ of the material of the beam are known, φ_B^e can be calculated.

a. Calculate the shape factor φ_B^e from the following experimental data, measured on an aluminum alloy beam loaded in 3-point bending (for which $C_1 = 48$; see Appendix B.3) using the data shown in the following table.

Attribute	Value
Beam stiffness S_B	7.2×10^5 N/m
Mass/unit length m/L	1 kg/m
Beam length L	1 m
Beam material	6061 aluminum alloy
Material density ρ	2670 kg/m^3
Material modulus E	69 GPa

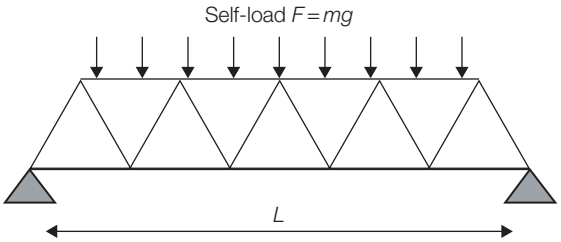


FIGURE E.23

b. A steel truss bridge shown in Figure E.23 has a span L and is simply supported at both ends. It weighs m tonnes. As a rule of thumb, bridges are designed with a stiffness S_B such that the central deflection δ of a span under its self-weight is less than 1/300 of the length L (thus $S_B \geq 300\, mg/L$ where g is the acceleration due to gravity, 9.81 m/s^2). Use the information listed in the table to calculate the minimum shape factor φ_B^e of the three steel truss bridge spans. Take the density ρ of steel to be 7900 kg/m^3 and its modulus E to be 205 GPa. The constant $C_1 = 384/5 = 76.8$ for uniformly distributed load (Appendix B.3).

Bridge and Date of Construction*	Span L (m)	Mass m (tonnes)
Royal Albert Bridge, Saltash, U.K. (1857)	139	1060
Carquinez Strait Bridge, CA (1927)	132	650
Chesapeake Bay Bridge, MD (1952)	146	850

* *Data from the* Bridges Handbook.

E9.5 Deriving indices for bending and torsion

a. A beam, loaded in bending, must support a specified bending moment M^* without failing and be as light as possible. Section shape is a variable, and "failure" here means the first onset of plasticity. Derive the material index. The following table summarizes the requirements.

Function	Lightweight beam
Constraints	Specified failure moment M^* Length L specified
Objective	Minimum mass m
Free variables	Choice of material Section shape and scale

b. A shaft of length L, loaded in torsion, must support a specified torque T^* without failing and be as cheap as possible. Section shape is a variable and "failure" again means the first onset of plasticity. Derive the material index. The table summarizes the requirements.

Function	Cheap shaft
Constraints	Specified failure torque T^* Length L specified
Objective	Minimum material cost C
Free variables	Choice of material Section shape and scale

E9.6 Use of the 4-quadrant chart for stiffness-limited design

a. Use the 4-quadrant chart for stiffness-limited design shown in Figure 9.9 to compare the mass per unit length m/L of a section with $EI = 10^5$ Nm^2 made from

1. structural steel with a shape factor φ_B^e of 20, modulus $E = 210$ GPa, and density $\rho = 7900$ kg/m^3
2. carbon-fiber–reinforced plastic with a shape factor φ_B^e of 10, modulus $E = 70$ GPa, and density $\rho = 1600$ kg/m^3
3. structural timber with a shape factor φ_B^e of 2, modulus $E = 9$ GPa, and density $\rho = 520$ kg/m^3.

The schematic in Figure E.24 illustrates the method.

b. Show, by direct calculation, that the conclusions of part (a) are consistent with the idea that to minimize mass for a given stiffness one should maximize $\sqrt{E^*/\rho^*}$ with $E^* = E/\varphi_B^e$ and $\rho^* = \rho/\varphi_B^e$.

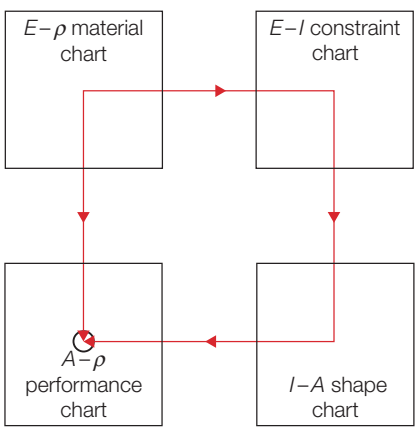


FIGURE E.24

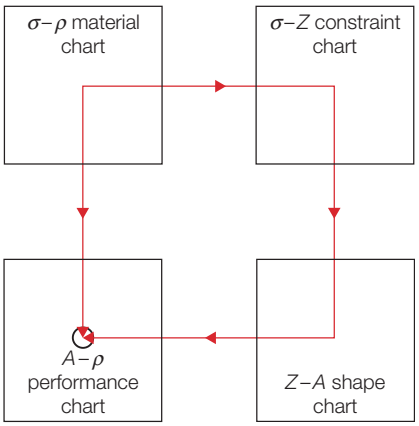


FIGURE E.25

E9.7 Use of the 4-quadrant chart for strength

a. Use the 4-quadrant chart for strength-limited design shown in Figure 9.12 to compare the mass per unit length m/L of a section with $Z_{\sigma f} = 10^4$ Nm (where Z is the section modulus) made from

1. mild steel with a shape factor φ_B^f of 10, strength $\sigma_f = 200$ MPa, and density $\rho = 7900$ kg/m^3
2. 6061-grade aluminum alloy with a shape factor φ_B^f of 3, strength $\sigma_f = 200$ MPa, and density $\rho = 2700$ kg/m^3
3. a titanium alloy with a shape factor φ_B^f of 10, strength $\sigma_f = 480$ MPa, and density $\rho = 4420$ kg/m^3

The schematic in Figure E.25 illustrates the method.

b. Show, by direct calculation, that the conclusions of part (a) are consistent with the idea that to minimize mass for a given stiffness one should maximize

$$\frac{(\varphi_B^f \sigma_f)^{2/3}}{\rho}$$

E9.8 A lightweight display stand Figure E.26 shows a concept for a lightweight display stand. The stalk must support a mass m of 100 kg, to be placed on its upper surface at a height h, without failing by elastic buckling. It is to be made of stock tubing and must be as light as possible. Use the methods in Chapter 9 to derive a material index for the tubular material of the stand that meets these requirements and that includes the shape of the section, described by the shape factor

$$\varphi_B^e = \frac{12\,I}{A^2}$$

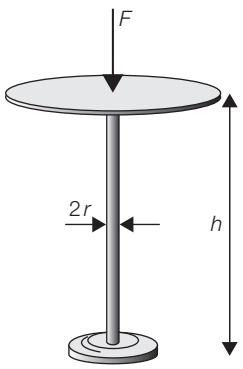


FIGURE E.26

where I is the second moment of area and A is the section area. The following table summarizes the requirements.

Function	Lightweight column
Constraints	Specified buckling load F Height h specified
Objective	Minimum mass m
Free variables	Choice of material Section shape and scale

Cylindrical tubing is available from stock in the following materials and sizes. Use this information and the material index to identify the best stock material for the column of the stand.

Material	Modulus E (GPa)	Tube Radius r	Wall Thickness/ Tube Radius t/r
Aluminum alloys	69	25 mm	0.07 to 0.25
Steel	210	30 mm	0.045 to 0.1
Copper alloys	120	20 mm	0.075 to 0.1
Polycarbonate (PC)	3.0	20 mm	0.15 to 0.3
Various woods	7–12	40 mm	Solid circular sections only

E9.9 Microscopic shape: Tube arrays Calculate the gain in bending efficiency ψ_B^e when a solid is formed into small, thin-walled tubes of radius r and wall thickness t that are then assembled and bonded into a large array, part of which is shown in Figure E.27. Let the solid of which the tubes are made have modulus E_s and density ρ_s. Express the result in terms of r and t.

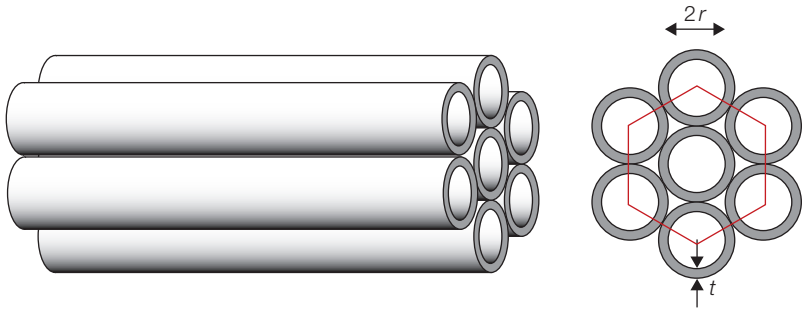


FIGURE E.27

E9.10 The structural efficiency of foamed panels (Figure E.28) Calculate the change in structural efficiency for both bending stiffness and strength when a solid flat panel of unit area and thickness t is foamed to give a foam panel of unit area and thickness h at constant mass. The modulus E and strength σ_f of foams scale with relative density ρ/ρ_s as

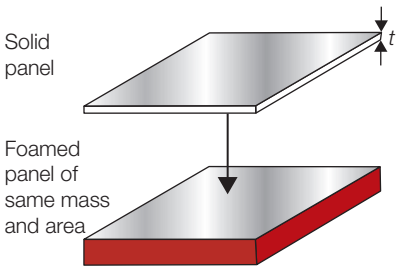


FIGURE E.28

$$E = \left(\frac{\rho}{\rho_s}\right)^2 E_s \quad \text{and} \quad \sigma_f = \left(\frac{\rho}{\rho_s}\right)^{3/2} \sigma_{f,s}$$

where E, σ_f, and ρ are the modulus, strength, and density of the foam and E_s, $\sigma_{f,s}$, and ρ_s those of the solid panel.

E.10 HYBRID MATERIALS (CHAPTERS 11 AND 12)

The examples in this section relate to the design of hybrid material described in Chapters 11 and 12. The first three involve the use of bounds for evaluating the potential of composite systems. The fourth is an example of hybrid design to fill holes in property space. Then comes one involving a sandwich panel. The next three make use of the charts for natural materials. The last is a challenge: to explore the potential of hybridizing two very different materials.

E10.1 Concepts for light, stiff composites Figure E.29 shows a chart for exploring stiff composites with light alloy or polymer matrices. A construction like that shown in Figure 11.7 of the text allows the potential of any given matrix-reinforcement combination to be assessed. Four matrix materials are shown, highlighted in red. The materials shown in gray are available as fibers (f), whiskers (w), or particles (p). The criteria of excellence (the indices E/ρ, $E^{1/2}/\rho$, and $E^{1/3}/\rho$ for light, stiff structures) are shown; they increase in value toward the top left. Use the chart to compare the performance of a titanium-matrix composite reinforced with (a) zirconium carbide, ZrC, (b) Saffil alumina fibers, and (c) nicalon silicon carbide fibers. Keep it simple: Use Equations (11.1) and (11.2) to calculate the density and upper and lower bounds for the modulus at a volume fraction of $f = 0.5$ and plot these points. Then sketch arcs of circles from the matrix to the reinforcement to pass through them. In making your judgment, assume that $f = 0.5$ is the maximum practical reinforcement level.

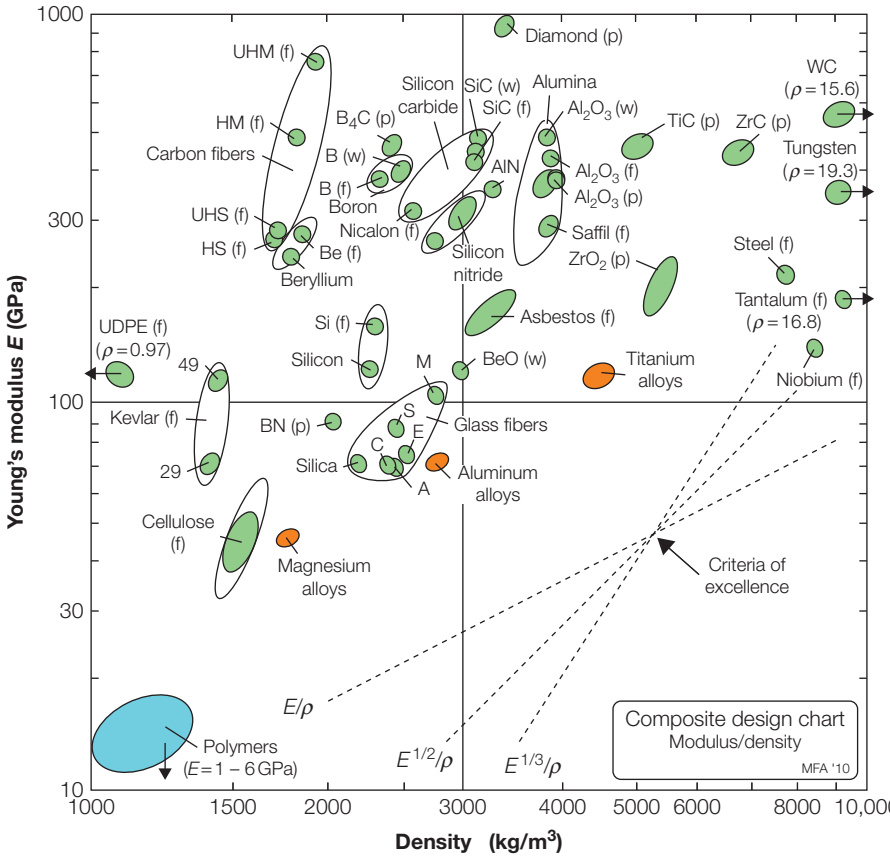


FIGURE E.29

E10.2 Use the chart shown in Figure E.29 to explore the relative potential of magnesium–E-glass-fiber composites and magnesium–beryllium composites for light, stiff structures.

E10.3 Concepts for composites with tailored thermal properties Figure E.30 shows a chart for exploring the design of composites with desired combinations of thermal conductivity and expansion, using light alloy or polymer matrices. A construction like that shown in Figure 11.10 allows the potential of any given matrix-reinforcement combination to be assessed. One criterion of excellence (the index for materials to minimize thermal distortion λ/α) is shown; it increases in value toward the bottom right. Use the chart to compare the performance of a magnesium AZ63 alloy-matrix composite reinforced with (a) alloy steel fibers, (b) silicon carbide fibers, SiC (f), and (c) diamond-structured carbon particles. Keep it simple: Use Equations (11.7) through (11.10) to calculate the upper and lower bounds for α and λ at a volume fraction of $f = 0.5$ and plot

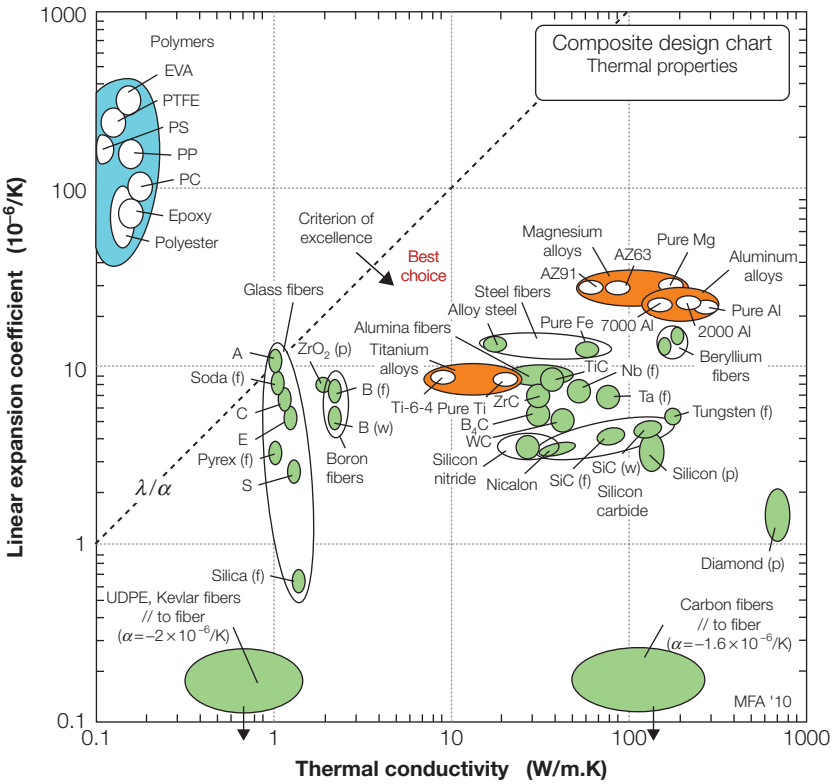


FIGURE E.30

these points. Then sketch curves linking matrix-to-reinforcement to pass through the outermost of the points. In making your judgment, assume that $f = 0.5$ is the maximum practical reinforcement level.

E10.4 Hybrids with exceptional combinations of stiffness and damping The loss-coefficient–modulus (η–E) chart shown in Figure 4.9 is populated only along one diagonal band. (The loss coefficient η measures the fraction of the elastic energy that is dissipated during a load-unload cycle.) Monolithic materials with low E have high η, those with high E have low η. The challenge here is to devise hybrids to fill the holes, with the following applications in mind.

a. Sheet steel (as used in car body panels, for instance) is prone to lightly damped flexural vibration. Devise a hybrid sheet that combines the high stiffness E of steel with high loss coefficient η.

b. High loss coefficient means that energy is dissipated on mechanical cycling. This energy appears as heat, sometimes with undesirable consequences. Devise a hybrid with low modulus E and low loss coefficient η.

E10.5 Sandwich panels An aircraft-quality sandwich panel has the characteristics listed in the following table.

a. Use the data and Equations (11.13) through (11.15) of the text to calculate the equivalent density, flexural modulus, and strength.
b. Plot these on a modulus–density and a modulus–strength chart. Do the flexural properties of the panel lie in a region of property space not filled by monolithic materials?

Data for Glass Fiber/Aluminum Honeycomb Sandwich Panel	
Face material	0.38-mm glass fiber/epoxy
Core material	3.2-mm cell, 97.8 kg/m^3, 5052 Alu honeycomb
Panel weight per unit area m_a	2.65 kg/m^2
Panel length L	510 mm
Panel width b	51 mm
Panel thickness d	10.0 mm
Flexural stiffness EI	67 N.m^2
Failure moment M_f	160 Nm

E10.6 Natural hybrids that are light, stiff, and strong

a. Plot aluminum alloys, steels, CFRP, and GFRP onto a copy of the E–ρ chart for natural materials (refer to Figure 12.13 in the text), where E is Young's modulus and ρ is the density. How do they compare, using the flexural stiffness index $E^{1/2}/\rho$ as the criterion of excellence?
b. Do the same thing for strength using a copy of Figure 12.14, using the flexural strength index $\sigma_f^{2/3}/\rho$ (where σ_f is the failure stress) as the criterion of excellence.

The following table lists the necessary data.

Material	Young's Modulus E (GPa)	Density ρ (kg/m^3)	Strength, σ_f (MPa)
Aluminum alloy	74	2700	335
Steel	210	7850	700
CFRP	100	1550	760
GFRP	21	1850	182

E10.7 Natural hybrids that act as springs The table that follows lists the moduli and strengths of spring materials. Plot these onto a copy of the E–σ_f chart for natural materials shown in Figure 12.15, and compare their

energy-storing performance with that of natural materials, using σ_f^2/E as the criterion of choice. Here σ_f is the failure stress, E is Young's modulus, and ρ is the density.

Material	Modulus (GPa)	Strength (MPa)	Density (kg/m^3)
Spring steel	206	1100	7850
Copper-2% beryllium	130	980	8250
CFRP filament wound	68	760	1580

E10.8 Finding a substitute for bone Find an engineering material that most closely resembles the longitudinal strength of compact bone (*cortical bone* (*L*)) in its strength/weight (σ_f/ρ) characteristics by plotting data for this material, read from Figure 12.14, onto a copy of the σ_f–ρ chart for engineering materials (refer to Figure 4.4). Here σ_f is the failure stress and ρ is the density.

E10.9 Creativity: What could you do with *x*? The same 68 materials appear on all of the charts in Chapter 4. These can be used as the starting point for "What if…?" exercises. As a challenge, use any chart or combination of charts to explore what might be possible by hybridizing any pair of the materials listed next, in any configuration you care to choose.

a. Cement
b. Wood
c. Polypropylene
d. Steel
e. Copper

E.11 SELECTING PROCESSES (CHAPTERS 13 AND 14)

The exercises in this section use the process selection charts in Chapters 13 and 14. They are useful in giving a feel for process attributes and the way in which process choice depends on material and shape. Here the CES software offers greater advantages: what is cumbersome and of limited resolution with the charts is easy with the software, which offers much greater resolution.

Each exercise has two parts, labeled (a) and (b). The first involves translation. The second uses the selection charts in Chapter 13 (which you are free to copy) in the way that was illustrated in Chapter 14.

E11.1 Elevator control quadrant The quadrant sketched in Figure E.31 is part of the control system for the wing elevator of a commercial aircraft. It is to be made of a light alloy (aluminum or magnesium) with the shape shown in the figure. It weighs about 5 kg. The minimum section thickness is 5 mm, and

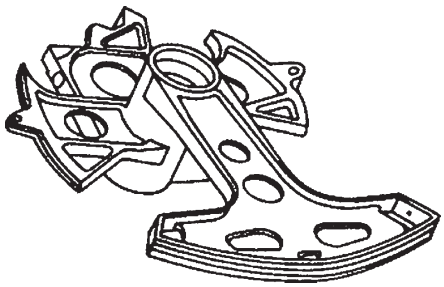


FIGURE E.31

—apart from the bearing surfaces—the requirements on surface finish and precision are not strict: surface finish $\leq 10\,\mu$m and precision ≤ 0.5 mm. The bearing surfaces require a surface finish $\leq 1\,\mu$m and a precision ≤ 0.05 mm. A production run of 100 to 200 is planned.

a. Itemize the function and constraints, leave the objective blank, and enter "Choice of process" for the free variable.
b. Use copies of the charts from Chapter 13 in succession to identify processes to shape the quadrant.
c. If the CES software is available, apply the constraints and identify in more detail the viable processes.

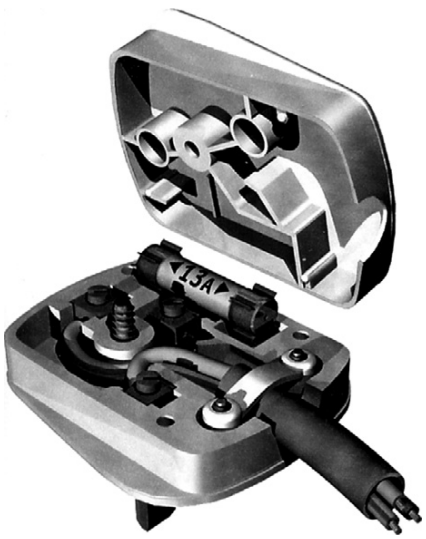


FIGURE E.32

E11.2 Casing for an electric plug The electric plug is perhaps the commonest of electrical products (Figure E.32). It has a number of components, each performing one or more functions. The most obvious are the casing and the pins, although there are many more (i.e., connectors; a cable clamp; fasteners; and, in some plugs, a fuse). The task is to investigate the processes for shaping the two-part insulating casing, the thinnest part of which is 2 mm thick. Each part weighs about 30 grams and is to be made in a single step from a thermoplastic or thermosetting polymer with a planned batch size of $5 \times 10^4 - 2 \times 10^6$. The required tolerance of

0.3 mm and surface roughness of 1 μm must be achieved without secondary operations.

a. Itemize the function and constraints, leave the objective blank, and enter "Choice of process" for the free variable.
b. Use the charts from Chapter 13 successively to identify possible processes to make the casing.

E11.3 Ceramic valves for taps Few things are more irritating than a dripping tap. Taps drip because the rubber washer is worn or the brass seat is pitted by corrosion, or both. Ceramics have good wear resistance, and they have excellent corrosion resistance in both pure and saltwater. Many household taps now use ceramic valves.

The sketch in Figure E.33 shows how they work. A ceramic valve consists of two disks mounted one above the other, spring-loaded so that their faces are in contact. Each disk has a diameter of 20 mm and a thickness of 3 mm, and weighs about 10 grams. In order to seal well, the mating surfaces of the two disks must be flat and smooth, requiring high levels of precision and surface finish; typically tolerance < 0.02 mm and surface roughness < 0.1 μm. The outer face of each has a slot that registers it and allows the upper disc to be rotated through 90° (a quarter-turn). In the off position the holes in the upper disc are blanked off by the solid part of the lower one; in the on position the holes are aligned. A production run of 10^5–10^6 is envisaged.

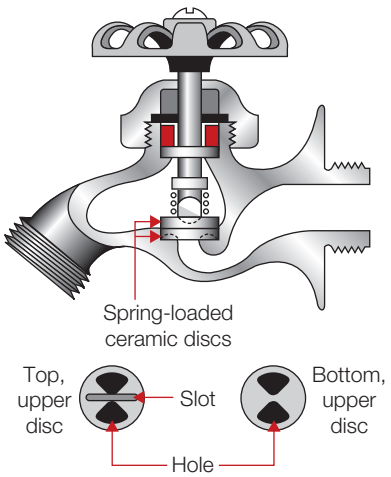


FIGURE E.33

a. List the function and constraints, leave the objective blank, and enter "Choice of process" for the free variable.
b. Use the charts from Chapter 13 to identify possible processes to make the casing.

E11.4 Shaping plastic bottles Plastic bottles are used to contain fluids as various as milk and engine oil (Figure E.34). A typical polyethylene bottle weighs about 30 grams and has a wall thickness of about 0.8 mm. The shape is 3D hollow. The batch size is large (1,000,000 bottles). What process could be used to make them?

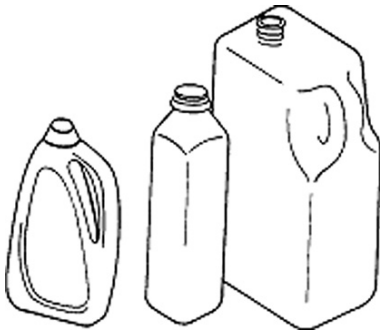


FIGURE E.34

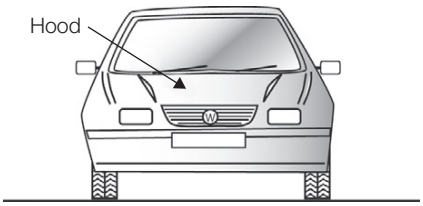


FIGURE E.35

a. List the function and constraints, leave the objective blank, and enter "Choice of process" for the free variable.
b. Use the charts from Chapter 13 to identify possible processes to make the casing.

E11.5 Car hood As weight saving assumes greater importance in automobile design, the replacement of steel parts with polymer-composite substitutes becomes increasingly attractive. Weight can be saved by replacing a steel hood with one made from a thermosetting composite (Figure E.35). The weight of the hood depends on the car model: A typical composite hood weighs is 8 to 10 kg. The shape is a dished sheet and the requirements on tolerance and roughness are 1 mm and 2 μm, respectively. A production run of 100,000 is envisaged.

a. List the function and constraints, leave the objective blank, and enter "Choice of process" for the free variable.
b. Use the charts from Chapter 13 to identify possible processes to make the casing.

E11.6 Complex structural channels Channel sections for window frames, for slide-together sections for versatile assembly, and for ducting for electrical wiring can be complex in shape. Figure E.36 shows an example. The order is for 10,000 such sections, each 1 m in length and weighing 1.2 kg, with a minimum section of 4 mm. A tolerance of 0.2 mm and a surface roughness of less than 1 μm must be achieved without any additional finishing operation.

a. List the function and constraints, leave the objective blank, and enter "Choice of process" for the free variable.
b. Use the charts from Chapter 11 to identify possible processes to make the casing.



FIGURE E.36

E11.7 Selecting joining processes This exercise and the next require the use of the CES software.

a. Use the software from CES to select a joining process to meet the following requirements.

Function	Create a permanent butt joint between steel plates
Constraints	Material class: carbon steel Joint geometry: butt joint Section thickness: 8 mm Permanent Watertight
Objective	–
Free variable	Choice of process

b. Use the software from CES to select a joining process to meet the following requirements.

Function	Create a watertight, demountable lap joint between glass and polymer
Constraints	Material class: glass and polymers Joint geometry: lap joint Section thickness: 4 mm Demountable Watertight
Objective	–
Free variable	Choice of process

E11.8 Selecting surface treatment processes This exercise, like the previous one, requires the use of the CES software.

a. Use CES to select a surface treatment process to meet the following requirements.

Function	Increase the surface hardness and wear resistance of a high carbon steel component
Constraints	Material class: carbon steel Purpose of treatment: increase surface hardness and wear resistance
Objective	–
Free variable	Choice of process

b. Use CES to select a surface treatment process to meet the following requirements.

Function	Apply color and pattern to the curved surface of a polymer molding
Constraints	Material class: thermoplastic Purpose of treatment: aesthetics, color Curved surface coverage: good or very good
Objective	–
Free variable	Choice of process

E.12 MATERIALS AND THE ENVIRONMENT (CHAPTER 15)

E12.1 Use the E–$H_p\rho$ chart shown in Figure 15.9 to find the polymer with a modulus E greater than 1 GPa and the lowest embodied energy per unit volume.

E12.2 A maker of polypropylene (PP) garden furniture is concerned that the competition is stealing part of his market by claiming that the "traditional" material for garden furniture, cast iron, is much less energy and CO_2-intensive than PP. A typical PP chair weighs 1.6 kg; one made of cast iron weighs 8.5 kg. Use the data for these two materials in Appendix A, Table A.10, to find out who is right; are the differences significant if the data for embodied energy are only accurate to ±20%?

If the PP chair lasts 5 years and the cast iron chair lasts 25 years, does the conclusion change?

E12.3 Identical casings for a power tool could be die-cast in aluminum or molded in ABS or polyester GFRP. Use the bar chart of embodied energy per unit volume shown in Figure 15.8 to decide which choice minimizes the material embodied energy, assuming the same volume of material is used for each casing.

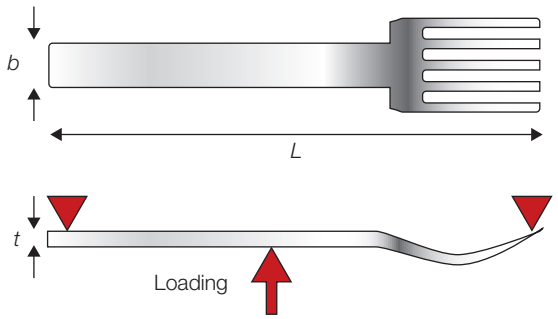


FIGURE E.37

E12.4 Disposable knives and forks Disposable knives and forks are ordered by an environmentally conscious pizza house (Figure E.37). The shape of each (and thus the length, width, and profile) are fixed, but the thickness is free: It is chosen to give enough bending stiffness to cut and impale the pizza without excessive flexure. The pizzeria proprietor wishes to enhance the greenness of her image by minimizing the energy content of the throw-away tableware, which could be molded from polystyrene (PS) or stamped from aluminum sheet.

Establish an appropriate material index for selecting materials for energy-economic forks. Model the eating implement as a beam of fixed length L and width w, but with a thickness t that is free, and loaded in bending, as in the figure. The objective function is the volume of material in the fork times its energy content $H_p\rho$ per unit volume (H_p is the embodied energy per kg, and ρ the density). The limit on flexure imposes a stiffness constraint (Appendix B.3). Use this information to develop the index.

Flexure in cutlery is an inconvenience. Failure—whether by plastic deformation or by fracture—is more serious: It causes loss of function; it might even cause hunger. Repeat the analysis, deriving an index when a required strength is the constraint.

E12.5 Show that the index for selecting materials for a strong panel with the dimensions shown in Figure E.38, loaded in bending, with minimum embodied energy content is that with the largest value of

$$M = \frac{\sigma_y^{1/2}}{H_p\rho}$$

where H_p is the embodied energy of the material, ρ its density, and σ_y its yield strength. To do so, rework the panel derivation in Chapter 5, Equation (5.9),

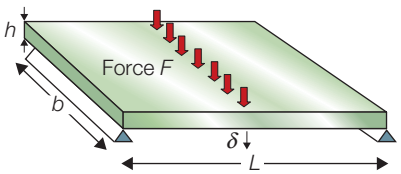


FIGURE E.38

replacing the stiffness constraint with a constraint on failure load F requiring that it exceed a chosen value F^* where

$$F = C_2 \frac{I\sigma_y}{hL} > F^*$$

where C_2 is a constant and I is the second moment of area of the panel $I = \frac{bh^2}{12}$.

E12.6 Use the indices for the crash barriers (Equations (15.9) and (15.10)) with the charts for strength and density (refer to Figure 4.4) and strength and embodied energy (refer to Figure 15.10) to select materials for each of the barriers. Position your selection line to include one metal for each. Reject ceramics and glass on the grounds of brittleness. List what you find for each barrier.

E12.7 The makers of a small electric car wish to make bumpers out of a molded thermoplastic. Which index is the one to guide this selection if the aim is to maximize the range for a given battery storage capacity? Plot it on the strength–density chart in Figure 4.4, and make a selection.

E12.8 Energy-efficient floor joists Floor joists are beams loaded in bending. They can be made of wood, of steel, or of steel-reinforced concrete, with the shape factors listed in the following table. For a given bending stiffness and strength, which of these carries the lowest production-energy burden? The relevant data, drawn from the tables in Appendix A, are listed.

a. Start with stiffness. Locate from Equation (9.20) in the text the material index for stiffness-limited, shaped beams of minimum mass. Adapt this to make the index for stiffness-limited, shaped beams of embodied energy by multiplying density ρ by embodied energy/kg H_p. Use the modified index to rank the three beams.

b. Repeat the procedure, this time for strength, creating the appropriate index for strength-limited shaped beams at minimum energy content by adapting Equation (9.28).

What do you conclude about the relative energy penalty of design with wood and with steel?

Material	Density ρ kg/m^3	Modulus E GPa	Strength σ_f MPa	Energy H_p MJ/kg	ϕ_B^e	ϕ_B^f
Soft wood	700	10	40	7.5	2	1.4
Reinforced concrete	2900	35	10	5	2	1.4
Steel	7900	210	200	30	15	4

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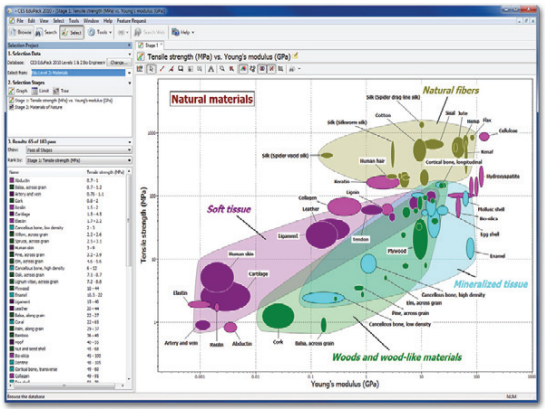



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